

Global Challenges and Initiatives in Kidney Health

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CHAPTER OUTLINE

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KEY POINTS

- The global burden of chronic kidney disease (CKD) is rising. In 2015, CKD was the 10th most common cause for death globally, directly causing 1 out of every 43 deaths. CKD contributes to a similar number of deaths from cardiovascular disease.
- CKD occurs in younger individuals in lower-income regions.
- The global burden of acute kidney injury (AKI) remains unknown.
- Kidney disease is a major risk factor for cardiovascular morbidity and mortality.
- Many countries lack policies regarding kidney disease advocacy, screening, and prevention.
- Achieving universal health coverage may reduce the global burden of AKI and CKD through improved access to primary care and essential medication.
- Access to dialysis and transplantation is highly inequitable across the globe and within regions and countries, largely dictated by ability to pay.
- If dialysis and transplantation are to be provided in low-resource settings, rigorous planning must be done initially to determine affordability and ensure sustainability, and transparent criteria must be developed to achieve fairness and equitable access to RRT.

GLOBAL CHALLENGES IN KIDNEY HEALTH

AWARENESS OF KIDNEY DISEASE AS AN IMPORTANT PUBLIC HEALTH ISSUE

Kidney disease is an important noncommunicable disease (NCD) worldwide; however, this is not yet widely acknowledged in the global policy agenda. The World Health organization (WHO) Global Action Plan for the Prevention and Control of NCDs has focused on addressing four main disease areas—heart diseases and stroke, cancer, chronic respiratory diseases, and diabetes, with the recent addition of mental health—and has identified these diseases as the major NCD killers globally.¹ According to WHO, in 2012, 46.2% of NCD deaths (17.5 million deaths) were attributed to cardiovascular disease (CVD), 21.7% (8.2 million deaths) to cancers, 10.7% (4 million deaths) to chronic respiratory diseases, and 4% (1.5 million deaths) due to diabetes, totaling 82% of all NCD deaths. As discussed in more detail later, estimates from the Global Burden of Disease (GBD) study in 2015 have shown that around 1.2 million people were known to have died of chronic kidney disease (CKD).² Others have estimated that at least 2.3 million people died in 2010 because they

had no access to dialysis, and 1.7 million die annually from acute kidney injury (AKI).^{3,4} It is possible, therefore, that kidney disease may contribute to more deaths than diabetes, respiratory diseases, or individual forms of cancer, which are among the four main NCDs targeted by the current NCD Action Plan.^{1,5} The global importance of kidney disease as a cause of mortality, as well as morbidity, has been increasing, and the Global Action Plan does acknowledge that synergies exist between NCDs, and that “conditions such as kidney disease result from lack of early detection and management of hypertension and diabetes, and therefore are closely linked to major non-communicable diseases.”¹

AKI and CKD are both of public health concern. Although their true prevalence in low-income countries (LICs) and lower middle-income countries (LMICs; collectively, LLMICs) is unknown, it is estimated that they are at least as high, if not higher, than in high-income countries (HICs).^{4,6,7} Approximately 750 million adults (8%–16%) worldwide are living with CKD, of whom around 78% reside in LLMICs.^{6,8} Up to 90% of those affected, however, are not aware of their condition.^{8,9} The prevalence of CKD among children is unknown but may reach 1% of the pediatric population.¹⁰ Traditional and universal risk factors for CKD include diabetes

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Table 75.1 Global Relevance of Major Risk Factors for Chronic Kidney Disease and Suggested Mitigation Strategies

Risk Factor	Global Prevalence	Primary Prevention	Projected Risk for CKD	Secondary Prevention of CKD		Knowledge Gaps	Relevance for HIC	Relevance for LIC	Advocacy Required	References
				Primary Prevention	Projected Risk for CKD	Secondary Prevention of CKD				
Diabetes type 2	All diabetes 387 million, with largest concentrations in Western Pacific (138 million) and Southeast Asia (75 million)	Education, lifestyle, diet, exercise, weight, management	~40% overall and >50% in most non-white populations	Glucose control, BP control, lifestyle factors (avoidance of high dietary protein), ACEI or ARB		Glucose targets, best medications, need for novel therapies for diabetic kidney disease	Obesity, DM, GDM	Increasing obesity and DM, GDM; poor facilities for diagnosis and treatment	Policy development around food content and prices of healthy food; urban planning to increase walking opportunities, tobacco; universal health care; access to diagnosis; reliable access to medication; lifestyle	127–130
Diabetes, type 1	Type 1—about 5% of overall global prevalence	Viral exposures (?)	~30 % Not known to vary by race/ethnicity	Glucose control, BP control, lifestyle factors (avoidance of high dietary protein), ACEI or ARB		Glucose targets, novel therapies for diabetic kidney disease	Glycemic control	Glycemic control; poor facilities for diagnosis and treatment	Universal health care, access to diagnosis, reliable access to medication, lifestyle	127
Hypertension	2010—31% of adults globally (28.5% in HICs, 31.5% LMICs); 1.39 billion people (349 million in HICs, 1.04 billion in LMICs)	Education, lifestyle, diet, exercise, weight, management, smoking, stress reduction	~10 %	Blood pressure control, ACEI or ARB if high-level albuminuria; other medication types?		BP targets; albuminuria-based?	Obesity, dietary sodium	Obesity, dietary sodium; strokes also; high awareness, Rx, and control vs. low in LMICs	Policy development around food sodium content, tobacco, alcohol; need to increase awareness; treatment and control globally; universal health care; consider polypill strategy; awareness, access to diagnosis, reliable access to medication, lifestyle	131

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Table 75.1 Global Relevance of Major Risk Factors for Chronic Kidney Disease and Suggested Mitigation Strategies (Cont'd)

Risk Factor	Global Prevalence	Primary Prevention	Projected Risk for CKD	Secondary Prevention of CKD		Knowledge Gaps	Relevance for LIC		Advocacy Required	References
				Prevention	Risk of CKD, optimal BMI and variance by race and ethnicity and age, safe and effective weight management strategies (e.g., bariatric surgery)		HIC	for LIC		
Obesity (risks may vary for childhood and adult obesity)	Adult: overweight, 2013: 36.9% men, 38.0% women. Obesity 2014: 10.8% men, 14.9% women Children: in 2014, 41 million children <5 years of age were overweight or obese (48% in Asia, 25% in Africa)	Education, lifestyle, diet, exercise, weight management, stress reduction	Unknown Proteinuria or macroalbuminuria present in 4%–10% obese patients In morbidly obese risk of GFR decline $\geq 30\%$ over 4 years was 48.2 per 1000 person years Adolescent obesity associated with HR of 6.9 for all ESKD and HR of 19.4 for diabetic ESKD	Diet, exercise, weight loss, bariatric surgery (HICs), ACEI, ARB for proteinuria	Risk of CKD, optimal BMI and variance by race and ethnicity and age, safe and effective weight management strategies (e.g., bariatric surgery)	Access to weight management programs	Access to weight management programs; social roots of obesity, poverty, culture; access to nutritious food	Policy development to regulate food content, food prices; urban planning to permit physical exercise; access to better diet; education, physical activity education		132–137
Medications (antibiotics, NSAIDs, PPI, counterfeit drugs, contrast media)	AKI: 24% globally related to nephrotoxins (29% HICs, 22% LMICs, 23% LLMICs) CKD: unknown	Improve awareness, prescription flagging, stop unnecessary prescriptions	70% of children with nephrotoxin-induced AKI had CKD at 6 months CKD risk variable, by medication	Early detection, urine screening for leukocytes, stop medications early	Burden of disease; which medication may increase risk for CKD	Electronic alert systems, prescription data-sharing databases, package warnings	Reduce counterfeit drugs; regulate drug manufacture to reduce adulterants	Awareness, prescription practices, marketing		24, 138, 139
Traditional, alternative remedies	Frequent use globally, >80% in LMICs	Improve awareness Improve access to alternatives (UHC)	35% of AKI in Africa; unknown contribution to CKD; increased risk of ESKD with consumption of some remedies	Stop medication, hydration	Burden of disease, toxic compounds	Huge market OTC and over the Internet, need regulation of the industry	Engage with communities to, for example, understand reasons for use, barriers to Western medicine	Policies to regulate manufacture and sale of alternative remedies; limit unfounded, fraudulent advertising; universal health care; awareness, collaboration with traditional healers, improve access to medical care and affordability of medication; encourage publication of case reports to build database		21, 140, 141

Kidney stones	Geographic variability Adults: 5%–9% Europe, 12% Canada, 13%–15% United States, 1%–5% East, 20% Saudi Arabia	Increase awareness of local risks; emphasize importance of hydration, certain infections	GFR tends to be reduced in stone formers vs. controls	Hydration, diet, recurrent stone prevention, early reversal of obstruction	Regional risks	High costs	Likely unrecognized important cause of CKD and infections	Access to clean water, reduce environmental and occupational risks; increase awareness of need for follow-up for CKD, CVD in stone formers	142–144
Low birth weight, SGA, prematurity	Globally: LBW 15%, preterm 10% In LMICs, 2010, 13.7 million babies preterm; 2010, 43.3 million babies LBW, SGA	Avoid obesity, healthy lifestyle	70% increased risk	Screen for BP and proteinuria; treat early	Would reduction affect future risk?	Increased maternal age, assisted reproduction, maternal chronic illness	Preeclampsia, maternal malnutrition, poverty, war, poor antenatal care, pregnancy spacing, child marriage	Awareness, public health measures, optimize maternal and child health, avoid childhood obesity, universal health care, document birth weights, prematurity in health record; need for long-term follow-up of children at risk	145, 146
Preeclampsia, eclampsia	2%–5% globally Global prevalence, 2013—1.3 million	Optimize maternal health pre-pregnancy	RR, HT, 3.7; RR, microalbuminuria, 4–8; RR, ESKD, 4.7; RR, kidney biopsy, 3.3	Screen for BP and proteinuria treat early	How to diagnose and prevent?	Prematurity, later CVD, ESKD	Prematurity, CVD, ESKD	Maternal health, access to antenatal care, universal health care; mothers require long-term follow-up for CKD and CVD	147–150
HIV	2013: 35 million worldwide, 24.7 million in sub-Saharan Africa	Education, condoms	Africa 6.0%–48.5%, Europe 3.5%, 18%, Hong Kong 18%, Brazil 1.1%–5.6%, India 27%, Iran 20%	PEP, HAART	Impact of HAART on all forms of renal disease, other kidney diseases in HIV-infected individuals	Competing risks of mortality	Poverty, suboptimal access to ART, ongoing infection risk	Policies around needle sharing, prostitution; universal health care; national policies for prevention education; access to ART, reduce gender and sexuality discrimination, empower women; surveillance of renal function on ART	151

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Table 75.1 Global Relevance of Major Risk Factors for Chronic Kidney Disease and Suggested Mitigation Strategies (Cont'd)

Risk Factor	Global Prevalence	Primary Prevention	Projected Risk for CKD	Secondary Prevention of CKD		Knowledge Gaps	Relevance for HIC	Relevance for LIC	Advocacy Required	References
				Treatment of hepatitis B	Treatment of hepatitis C					
Hepatitis B	Global prevalence 2013—331.0 million	Education, reduce scarification, vaccination	Hepatitis B-associated GN, 3% France, 3% China			Impact of routine vaccination on CKD burden	Reduce HCC, liver failure, transplantation	High prevalence	Policies around needle sharing, vaccination; advocacy for sexual health, drug abuse; equity in access to vaccination (vaccination reduced membranous nephropathy among Taiwanese children); equity in access to antiviral therapy	150, 152, 153
Hepatitis C	Global prevalence 2013: 147.8 million	Education	Global: 10%–16%; glomerular lesions in 54.8% of HCV-positive subjects at autopsy	Treatment of hepatitis C		Impact of treatment on disease burden	Reduce HCC, liver failure, kidney transplantation; new medication very costly	Lower prevalence, unlikely to gain access to expensive therapies	Policies around needle sharing; advocacy against drug abuse; lobby for access to therapy in HICs and LMICs	150, 152, 154
Bacterial skin diseases	Global prevalence 2013: 5–8 million	Sanitation, early treatment	Acute PSGN, 9/100,000 in LMICs Higher frequency of CKD after poststreptococcal GN, worse in adults	Early detection of renal involvement; treatment and follow-up		Contribution to CKD burden in LICs unknown	Likely low	Likely high	Policies to improve child nutrition, school feeding programs; poverty, overcrowding; scabies prevention and early treatment; consider screening school children for hematuria, proteinuria	150, 155, 156
Schistosomiasis	Global prevalence, 2013: 290.6 million	Safe water Education	Obstruction, 2%–62%, chronic glomerulonephritis (hepatosplenic) in 15%	Prompt treatment; screening for obstruction		Obstruction usually not severe, renal function preserved; regional contribution to ESKD may be 3%–7% (Egypt)	Low	High regional	Public health policies, neglected tropical diseases; clean water; consider screening school children for hematuria, proteinuria; prompt access to diagnosis and treatment	150, 157–161

Diarrheal illnesses	Global prevalence 2013: 4.24 million	Safe water Sanitation, nutrition, vaccination	Important cause of AKI worldwide	Appropriate hydration, antibiotics when needed	Burden of CKD related to diarrhea-associated AKI, impact of vaccination on AKI, CKD	Relatively low, diarrhea-associated HUS	High, important cause of childhood AKI through volume depletion, sepsis, HUS	Public health policies, sanitation, water education, infrastructure, vaccination; advocacy to chlorinate water; handwashing, improve water safety, equitable access to vaccination, education about oral rehydration therapy	150
Malaria	Worldwide prevalence 2013: 351 million	Use of ITNs Vector control Prompt treatment with correct drugs	AKI <1%—>50% in adults with severe <i>Falciparum</i> malaria; CKD not often reported among survivors	Early screening and diagnosis and management	Contribution to CKD burden regionally unknown; possible differences among those living in endemic areas (or not?); may be associated with CKDu	Low	High regional	Public health policies: vector control, insecticide-treated nets, monitor medication resistance, combat counterfeit medication, introduce RDT; access to prevention, diagnosis, appropriate treatment Public health policies 30, 150, 162–164	
Tuberculosis	Worldwide prevalence 2013: 12.1 million	Healthy diet, reduce poverty, reduce HIV	Genitourinary, 27% of extrapulmonary TB (obstruction, parenchymal infection, interstitial nephritis)	Prompt diagnosis and full treatment	Low	Low, higher in immigrants, prison, indigenous, immunosuppressed populations	High, regional; often coinfection with HIV	Public health policies 150, 165 about detection, supervision of therapy, GeneXpert System, management of MDR, XDR, integration with HIV services; poverty, comorbid illness, nutrition, overcrowding, occupational exposure (mining), HIV infection	

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Table 75.1 Global Relevance of Major Risk Factors for Chronic Kidney Disease and Suggested Mitigation Strategies (Cont'd)

Risk Factor	Global Prevalence	Primary Prevention	Projected Risk for CKD	Secondary Prevention of CKD		Knowledge Gaps	Relevance for HIC	Relevance for LIC	Advocacy Required	References
Leptospirosis	Global incidence: 1.03 million	Use of ITNs, vector control, prompt treatment	AKI (Weil disease) 10%–60%	Early diagnosis	Contribution to burden of CKD unknown	Little	High, regional	Public health policies, neglected tropical diseases; poverty, water quality, overcrowding		30, 166
Environmental factors	(?) Risk factor prevalence for CKDu—likely association with environment, occupation, poor fluid intake, coinfections, traditional remedies	Avoid occupation, climate, environmental hazards	Prevalence: 13%–26% in high-risk populations	Hydration; avoid nephrotoxins	Causes and pathophysiology unknown	Low	CKDu major problem in multiple LMICs	Policies around working conditions, environmental contamination		27, 164, 167, 168
AKI	21% of hospital admissions (global data insufficient for accurate quantitation)	Early risk identification Treat underlying cause early Avoid nephrotoxins	Adults: 25.8/100 person-years (CKD), 8.6/100 person-years (ESKD) Children: 3.1/100 person-years (proteinuria), 0.9/100 person-years (ESKD)	Early diagnosis and treatment of AKI	Actual risk of CKD after AKI in population, impact of interventions to reduce AKI on CKD prevalence	Predominantly hospital-acquired, older adults, multiple comorbidities	Predominantly community-acquired, adults younger, few comorbidities	Increase awareness of risk of AKI and need for prompt treatment; require accessible methods to diagnose AKI; awareness of risk of CKD requiring long-term follow-up after severe AKI		4, 25, 26

ACEI, Angiotensin-converting enzyme inhibitor; AKI, acute kidney injury; ARB, angiotensin receptor blocker; ART, antiretroviral therapy; BMI, body mass index; BP, blood pressure; CKD, chronic kidney disease; CKDu, chronic kidney disease of uncertain cause; CVD, cardiovascular disease; DM, diabetes mellitus; GDM, gestational diabetes mellitus; GFR, glomerular filtration rate; GN, glomerulonephritis; ESKD, end-stage kidney disease; HAART, highly active antiretroviral therapy; HIC, high-income country; HIV, human immunodeficiency virus; HCV, hepatitis C Virus; ITN, insecticide-treated nets; LBW, low birth weight; LMIC, low middle-income country; MDR, multidrug resistance; NSAID, nonsteroidal antiinflammatory drug; OTC, over-the-counter; PEP, postexposure prophylaxis; PPI, proton pump inhibitor; PSGN, poststreptococcal glomerulonephritis; Rx, treatment; SGA, small for gestational age; TB, tuberculosis; UHC, universal health care; UMIC, upper middle-income country; XDR, extensive drug resistance.

Adapted from Luyckx VA, Tuttle KR, Garcia G, et al. Reducing major risk factors for chronic kidney disease. *Kidney Int Suppl* 2017;7:71–87.

and hypertension (Table 75.1).¹¹ Nontraditional risk factors for CKD are being increasingly recognized, including nephrotoxins (e.g., prescription medication and complementary remedies), kidney stones, fetal and maternal exposures, infections, environmental and occupational exposures, and AKI (see Table 75.1).^{7,12,13} For example, the use of traditional remedies is widespread, and some are known to be nephrotoxic.¹⁴ The contribution of such nontraditional risk factors to the burdens of AKI and CKD is unknown but may be significant, especially in LLMICs.

The public health importance of CKD lies not only in the risk of end-stage kidney disease (ESKD) requiring dialysis or transplantation, at great expense to the individual and the health system, but also because CKD is a major risk multiplier of cardiovascular risk, contributing to excess strokes and heart attacks and leading to death of many CKD patients before they ever reach ESKD.^{15–17} It is frequently underappreciated that CKD alone is a stronger risk factor for coronary events than diabetes alone and, when the two conditions coexist (which occurs in one of every three patients with diabetes), the risk of cardiovascular events and overall mortality is further multiplied.¹⁶

AKI occurs in 7% to 21% of hospitalized patients (30%–70% of critically ill patients) worldwide, affecting around 13.3 million people/year, of whom around 85% reside in LLMICs.^{4,18,19} Data for children are again relatively scarce, with estimates of 7.5% to 23.3% among hospitalized children, depending on the definition used.²⁰ Overall mortality has been estimated at 11% to 23% in adults and 14% in children, but varies with the population studied (up to 80% if no dialysis is available when required).^{4,19,21} AKI is associated with high costs of care because many patients require intensive care management, and mortality is lower in countries with higher expenditures on health care, reflecting better access to health care and dialysis in these countries.¹⁹ In higher-income countries, most cases of AKI are hospital-acquired, whereas in many LLMICs, community-acquired AKI is more common.^{18,22,23} Sepsis, dehydration, use of nephrotoxins, and primary AKI are major universal causes of AKI.²⁴ In many patients with AKI, kidney function will recover, but adults and children experiencing AKI have up to an eightfold risk of developing CKD and are at increased risk of longer-term CVD and death.^{12,25,26} CKD, in turn, is an important risk factor for AKI.¹²

Challenges to diagnosis of kidney disease include the necessity for access to blood and urine testing, awareness among health care workers, a degree of therapeutic nihilism in environments where access to specialist care and dialysis is limited, and the fact that kidney disease is often asymptomatic or symptoms are nonspecific until advanced stages. Proactive screening for early detection of AKI or CKD in high-risk individuals to permit interventions to delay worsening of the disease is therefore required.^{8,27–29} Increased awareness of risk factors for CKD and AKI in communities and hospitalized populations is urgently needed to implement prevention strategies successfully.¹² Kidney disease also bridges communicable and NCDs. Acute infections such as malaria and leptospirosis may lead to AKI, and chronic infections such as hepatitis B and C and HIV are important causes of CKD.^{12,13,30} In turn, AKI and CKD increase the risk and morbidity of infections.³¹ Many people worldwide are therefore affected by kidney disease, at least some of which is

preventable. Despite the best efforts at prevention, however, some people will develop either acute or end-stage kidney failure and require renal replacement therapy (RRT)—dialysis or transplantation—to remain alive.

RENAL REPLACEMENT THERAPY AND THE CHALLENGES OF LIMITED ACCESS

Many barriers reduce access to dialysis around the globe.^{3,21,32–35} Unless dialysis is covered through state coverage or insurance schemes, access to dialysis is often restricted by prohibitive out-of-pocket (OOP) expenditures, patient age or gender, geography, and lack of expertise and infrastructure (Fig. 75.1).^{36,37}

The global incidence of ESKD in 2010 was 93 (95% confidence interval [CI], 90–95) per million population (pmp), but varies significantly across the globe (Fig. 75.2A).^{11,17} The true incidence of ESKD in LLMICs is unknown because existing data come only from those who have access to dialysis and where data exist. The incidence of ESKD in LLMICs is often higher than the prevalence, indicating high attrition rates, predominantly because of the inability to pay for dialysis or being too sick at presentation.¹¹ In 2010, the global average prevalence rate of ESKD was 284 pmp; however, individual country prevalence ranged from 0 to over 2000 pmp, reflecting large global inequities (see Fig. 75.2B).¹⁷ Among children, the global prevalence rate is estimated to be under 100 pmp, but again is highly variable, depending on the accessibility of chronic dialysis.^{10,35,38,39} It has been estimated from the literature and available registries that there are large treatment gaps between those who need and those who obtain dialysis globally, with the largest gaps in Africa (84%–91% not treated) and Asia (66%–83% not treated).³ It has been estimated that at least 2.3 million people died prematurely in 2010 because they could not access dialysis.³ These deaths occurred predominantly in LLMICs. Other authors have estimated that a total of 1.2 million premature deaths resulted from lack of access to RRT among people with hypertension and diabetes alone, based on prevalence and expected rates of ESKD.⁴⁰ Of major concern is the prediction that by 2030, attributable to population aging and increasing rates of diabetes and hypertension, the global need for dialysis will double, with the greatest growth occurring in LLMICs.³

Few data are available regarding the need for dialysis for AKI in LLMICs. In general, AKI appears more severe at diagnosis in LLMICs compared with HICs, most likely due to delays in presentation to hospital, and therefore the need for dialysis is higher, but access to dialysis is lower in LLMICs.^{4,21,24} Acute peritoneal dialysis (PD) is an attractive modality for AKI because this is as effective as hemodialysis (HD), requires far less infrastructure, and can be performed with solutions and catheters adapted to local resources.^{41,42} PD should be cheaper than HD, especially if solutions are manufactured locally.^{43,44} However, owing to the necessity to import fluids, customs, and corruption, the costs of 5 days of PD or HD for a child in Nigeria, for example, were similar, and amounted to several times the monthly minimum wage, making OOP payment impossible for most.^{45,46} In recent years, some governments (e.g., Ghana, Tanzania, Nigeria) have introduced coverage for short-term dialysis for AKI paid for by national health insurance schemes, inpatient government-funded services, or nongovernmental organizations and charities.^{47–50} If renal

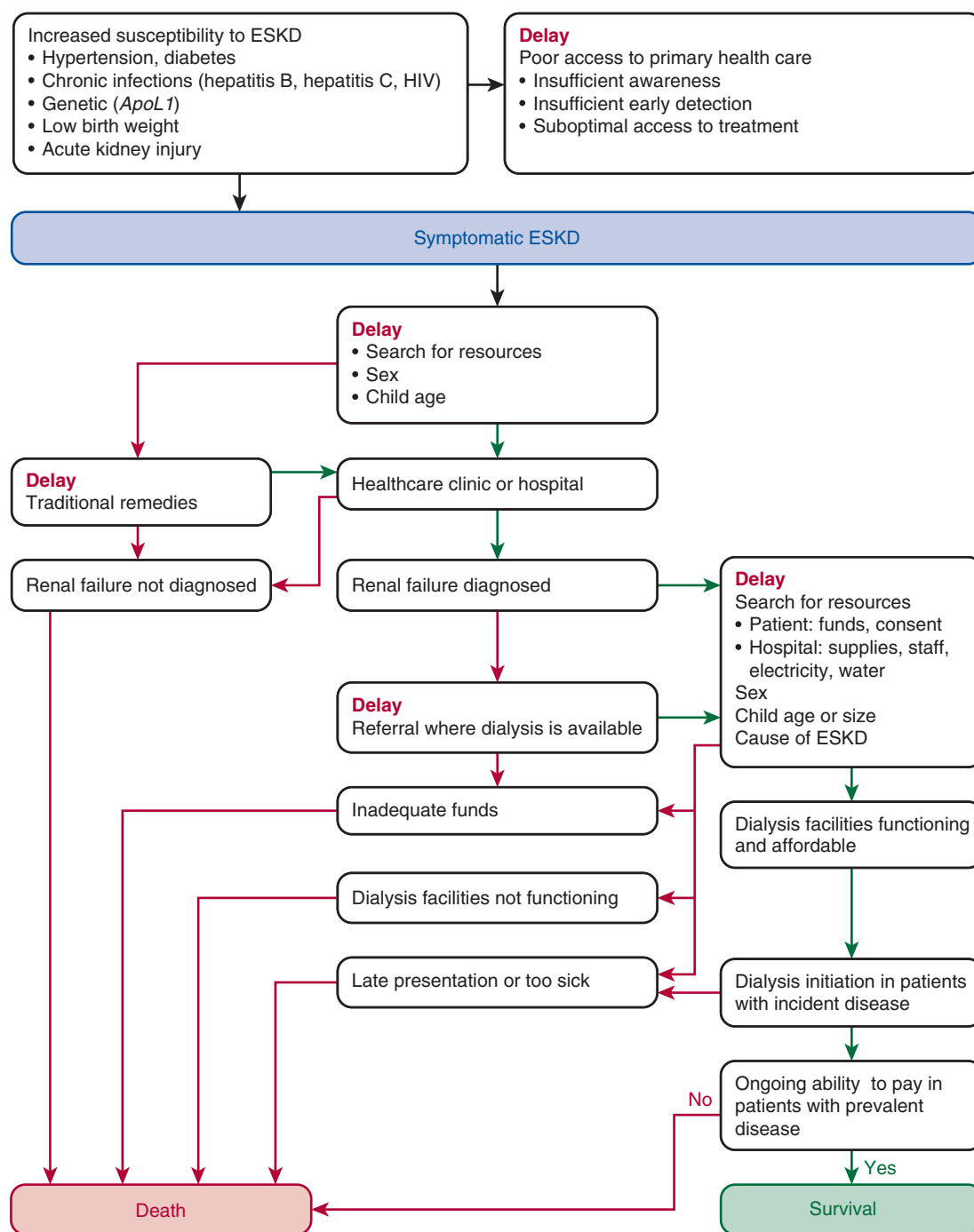


Fig. 75.1 Barriers to care in end-stage kidney disease (ESKD). This flow diagram shows barriers to access to dialysis contributing to mortality in ESKD in many sub-Saharan African countries. *Green arrows* show factors present to facilitate diagnosis of ESKD or referral for, or access to dialysis; *red arrows* show absence of these factors. Most barriers are related to access to care, access to diagnosis, out-of-pocket payments needed, and infrastructural resources. *ApoL1*, Apolipoprotein L1. (Adapted from Ashuntantang G, Osafo C, Olowu WA, et al. Outcomes in adults and children with end-stage kidney disease requiring dialysis in sub-Saharan Africa: a systematic review. *Lancet Glob Health*. 2017;5[4]:e408–e417.)

recovery does not occur within the coverage period, however, costs revert to OOP, which are unsustainable, and dialysis must then be actively discontinued.^{47,48,50}

Transplantation is the most cost-effective RRT solution in the long term; however, again, if costs of the surgery and long-term medication and follow-up are not sustainably covered by governments, health insurance, or donors, accessibility and success are significantly reduced.^{29,51,52} In India, for example,

72% of patients undergoing kidney transplantation reported some form of resulting financial crisis, with 10% expressing regret at having embarked on the procedure.⁵³ Globally, most transplantations are conducted in HICs, in part due to lack of resources and knowledge in LICs, as well as cultural practices and absence of legal frameworks governing organ donation (Fig. 75.3).⁵² Worldwide, transplantation activity is directly associated with health expenditure, physicians per

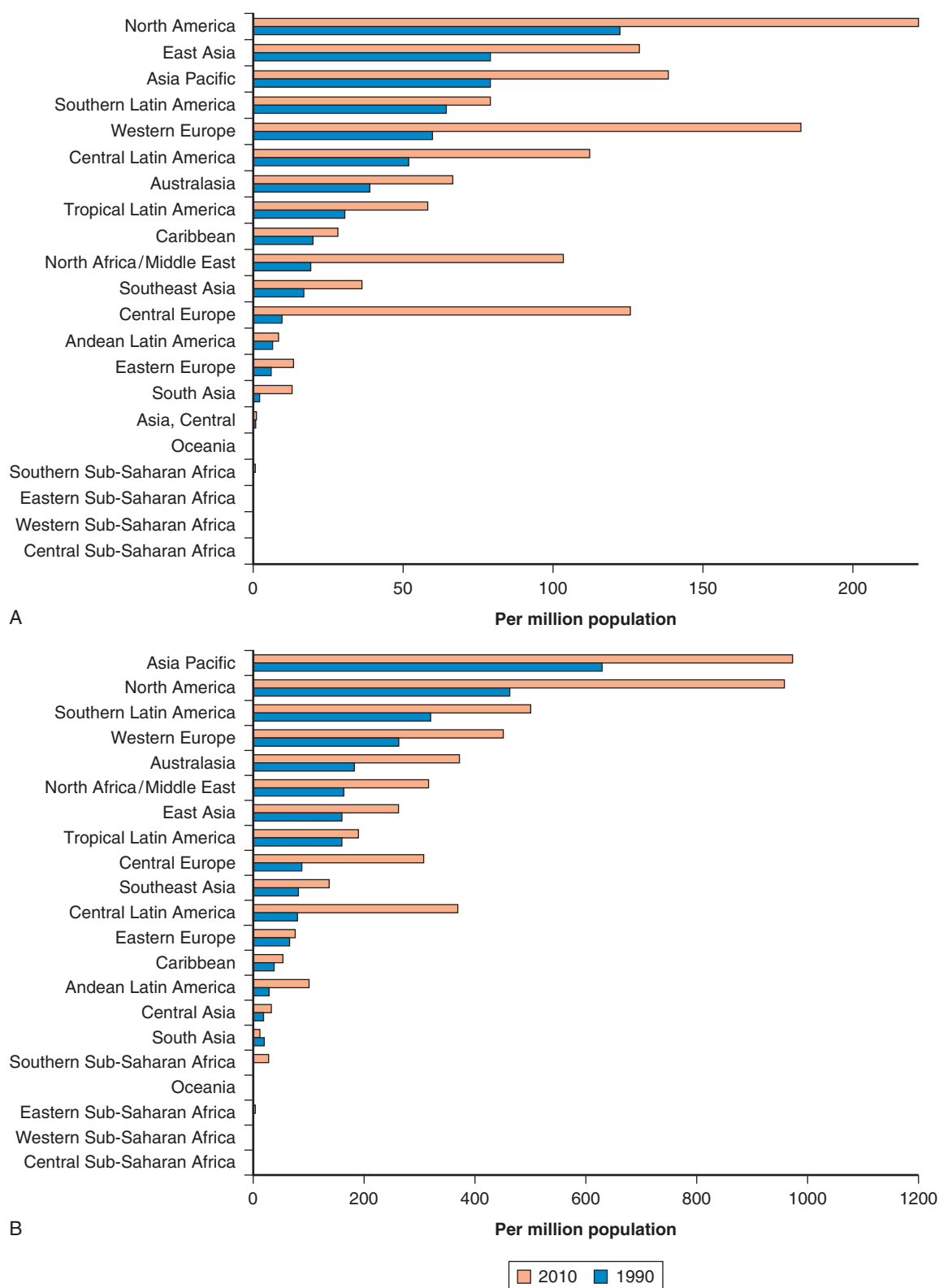


Fig. 75.2 Age-standardized maintenance dialysis. (A) Incidence/million population and (B) prevalence/million population for 21 world regions in 1990–2010. (From Thomas B, Wulf S, Bikbov B, et al., Maintenance dialysis throughout the world in years 1990 and 2010. *J Am Soc Nephrol.* 2015;26[11]:2621–2633.)

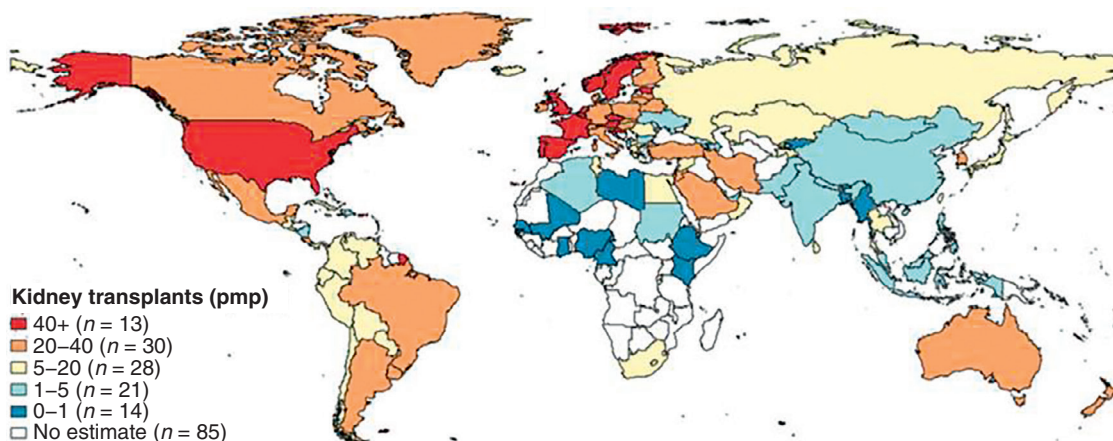


Fig. 75.3 Global distribution of kidney transplantation rates/million population (*pmp*) in 2012. (From Muralidharan A, White S. The need for kidney transplantation in low- and middle-income countries in 2012: an epidemiological perspective. *Transplantation*. 2015;99[3]:476–481.)

capita, and life expectancy at birth and inversely associated with reliance on OOP payments and reliance on external resources for health care.⁵² In some countries where the local population cannot access kidney transplantation, patients and donors travel abroad, if resources exist, for transplantation.^{54,55} Organ trafficking and commercial transplantation are a global problem, not only because of the ethical dilemmas they pose, but also because they are associated with worse outcomes for the transplant recipient. Also, they put living donors at risk in low-income environments, where they are unlikely to have been well screened and cannot access routine follow-up care.^{56,57}

DILEMMAS POSED BY LACK OF SUSTAINABLE RENAL REPLACEMENT THERAPY IN LOW-RESOURCE SETTINGS

Attempting to meet dialysis costs OOP generally constitutes a catastrophic health expenditure for a patient and family. Protection from individual financial risk is one of the cornerstones of the global push for universal health coverage; however, how far a government should go to meet the needs of patients requiring dialysis, potentially at the expense of funding other programs, is a contentious issue.^{58,59} The limited health budgets and current burdens of renal and nonrenal disease in LLMICs make progress toward universal access to RRT challenging.⁶⁰ Globally, on average, 0.15% of the population requires treatment for ESKD, at a cost of around 2% to 4 % of health expenditures.^{9,29} These direct costs of ESKD often do not account for costs of frequent hospitalizations and management of other comorbidities; therefore, total costs are even higher.^{9,29} Such disproportionate expenditures on a relatively small proportion of the population pose a challenge in terms of prioritization of health expenditure and opportunity costs. Some governments (e.g., Guinea, Cameroon, Senegal, India), alone or in partnership with private institutions or charities, have elected to subsidize the cost of chronic dialysis, requiring OOP payments of a few dollars a day.^{61–63} Even these prices, which are much lower than the average unsubsidized cost of \$60 to \$100 for a single HD treatment, remain unsustainable for families over the long term and often do not cover the vital costs of transport,

medications, or time lost from work, which can almost double the total OOP cost for a patient.^{21,35,64–68} In sub-Saharan Africa and India, most patients who initiate chronic dialysis cannot sustain it and stop dialysis within days to weeks.^{33,35} Even among European cohorts, mortality of children on dialysis was related to public health expenditures.⁶⁹ Sustainable and sufficient funding is therefore necessary for the successful treatment of kidney failure. It is unrealistic to imagine that in the short to medium term, all those requiring RRT will be able to access it equitably. Recognition of this fact is important to ensure that patients are still treated with dignity and have universal access to symptom relief and palliative care.

Clinical Relevance

Catastrophic Health Expenditures

Out-of-pocket health expenditure amount to >10% of total household expenditures or 40% of household nonfood expenditures.⁵³ Such expenses risk plunging families into poverty and reduce their capacity to obtain other necessities.

Clinical Relevance

Universal Health Coverage (UHC)

This is defined by WHO as “all people receiving quality health services that meet their needs without being exposed to financial hardship in paying for the services.”⁵⁹ The three dimensions of UHC include expanding priority services, expanding the number of people covered, and reducing out-of-pocket payments and thereby financial hardship.

Clinical Relevance

Opportunity Costs

These are health gains that could have been realized had the money been allocated to an alternative intervention.²⁹

POLICY DEVELOPMENT AND ACCESS TO RENAL REPLACEMENT THERAPY

Most states in LLMICs cannot afford to provide universal access to RRT; however, if states choose to provide some RRT, whether through sole funding or through partnership agreements, it is important that transparent and just eligibility criteria governing who will gain access to RRT are established.⁵⁸ Development of such criteria should facilitate bedside discussions and decision making and likely will involve consideration of clinical, social, and environmental realities that may affect a patient's ability to continue dialysis without incurring catastrophic health expenditures. Official criteria will also reduce moral distress among health care workers, removing the burden of individual case-by-case, life-and-death decision making.⁷⁰ Eligibility criteria may be more flexible for AKI than ESKD because the personal resources required will be less for AKI. To be effective, however, the development of such guidelines requires a transparent, consultative, and iterative development process and engagement with many different stakeholder groups. Potential policy scenarios and processes for development of dialysis guidelines are outlined elsewhere.^{58,71}

Policy options would include the following: (1) provision of RRT to all; (2) provision of dialysis for AKI only; (3) provision of RRT to a restricted eligible group (i.e., rationing); and (4) no state provision of RRT and allowing market forces to determine access (Table 75.2). Major inequities are inherent in each of these scenarios:

- Where universal RRT is not covered, inequities exist between countries.
- Where AKI only is covered, inequities exist between those with AKI and those with ESKD.
- Where rationing occurs, inequities exist between those meeting and not meeting the eligibility criteria.
- Where dialysis is not covered by the state, inequities exist within countries and regions depending on ability to pay and location.

In addition, when developing RRT policies and guidelines, consideration must also be given to achievable minimum safe standards of practice realistically because patients and providers may attempt to reduce dialysis frequency and dose to curb costs at the expense of health.³⁵

It is also important to recognize, however, that other inequities and health challenges in many LLMICs may be more urgent and affect many more people than kidney disease. In fact, the WHO specifically advises against the provision of dialysis in countries that have not yet met the needs of expanding priority health services (e.g., vaccinations, treatment of diarrhea, pneumonia, maternal health, hypertension, diabetes; http://www.who.int/choice/documents/making_fair_choices/en). Individual financial risk protection is, however, one of the three dimensions of universal health coverage; therefore, dialysis provision poses a dilemma—people die without it, people are impoverished because of it, and opportunity costs are extremely high. Priority setting within each country is therefore essential but requires reliable relative assessments of the disease burden, which is lacking for kidney disease in most LLMICs. Chronic dialysis for ESKD is suggested by some to be an unacceptable trade-off given that in Kenya, for example, 300 times more healthy life-years could be saved if dialysis expenditures were spent on tuberculosis control instead.⁵⁹ Once a country has made significant progress toward universal health coverage, however, the provision of acute dialysis for AKI is seen as cost-effective.⁵⁹ Whichever policies are put in place, it is crucially important to advocate for the need for prevention, early detection, and treatment of AKI and CKD as the most effective means to reduce health expenditures and the need for RRT.

GLOBAL INITIATIVES FOR KIDNEY HEALTH

In recent years, various initiatives have been launched or supported by the global nephrology community to improve understanding of the burden of kidney disease, estimate the

Table 75.2 Simplified Ethical Analysis of Potential Government Policies for Renal Replacement Therapy

Policy	Expected Health Benefits for Patients	Potential Harm and Burdens	Impact on Autonomy	Impact on Equity	Expected Efficiency
Universal coverage of dialysis for AKI and ESKD	All benefit	High opportunity costs	Good	Good	Expensive
Universal coverage for AKI only	AKI patients benefit	What if AKI does not recover?	Good for AKI	ESKD not considered	Less expensive
State coverage under limited conditions (rationing, AKI and ESKD)	Eligible patients benefit	Ineligible patients die; perception of unfairness	Need rationing	Inequity within patients with AKI and ESKD	Less expensive
No state coverage	Limited to rich few	Much harm, especially if problem “ignored” and no official guidelines; moral distress for health care workers needing to decide; may exacerbate brain drain	Very restricted	Very inequitable	Left to the market

AKI, Acute kidney injury; ESKD, end-stage kidney disease.

Adapted from Luyckx VA, Miljeteig I, Ejigu AM, et al. Ethical challenges in the provision of dialysis in resource-constrained environments. *Semin Nephrol.* 2017;37(3):273–286.

resources available to tackle kidney disease, advocate for kidney health, and contribute to the development of innovative strategies to prevent or detect and treat kidney disease early. Some of these initiatives are described later.

GLOBAL BURDEN OF DISEASE STUDY

MORTALITY AND MORBIDITY DUE TO CHRONIC KIDNEY DISEASE

The most comprehensive assessment of the global epidemiology of CKD has been carried out within the framework of the GBD study, which has described CKD mortality and morbidity in 195 countries from 1990 to 2015 (for yearly updated data please see <https://vizhub.healthdata.org/gbd-compare>). The combined CKD data from the GBD study are summarized in

Table 75.3. According to the latest published evidence,^{72–74} global CKD stages 3 to 5 prevalence was 322.5 million (4.6% of the world's population) in 2015. This estimate is higher, or at least comparable, with the prevalence of ischemic heart disease, chronic obstructive pulmonary disease, osteoarthritis, malaria, or the number of people living with the consequences of road injury.⁷²

Most cases of CKD stages 3 to 5 in 2015 were assumed to be due to diabetic nephropathy (100.8 million), followed by hypertensive nephropathy (79.0 million) and chronic glomerulonephritis (67.3 million). Notably, as many as 94.6 million people had CKD due to other causes, including unknown causes. Thus, 52.6% of CKD cases were attributed to diabetes or hypertension, whereas almost the same number of people had intrinsic kidney disease. CKD was more

Table 75.3 Global Burden of Low GFR (Chronic Kidney Disease [CKD] Stages 3–5)

Parameter	CKD Stages 3–5	Low GFR Overall ^a	CVD Due to Low GFR
Prevalence			
Number, 2015	322.5 million	—	—
Crude rate/100,000 population, 2015	4375.4	—	—
Change in number from 1990 to 2015, %	78.3	—	—
Change in crude rate from 1990 to 2015, %	28.3	—	—
Change in age-standardized rate from 1990 to 2015, %	0.4	—	—
Years Lived With Disability			
Number, 2015	8.2 million	9.3 million	1.0 million
Crude rate, 2015, per 100,000 population	110.9	126.7	13.4
Global rank, 2015	24th	11th	—
Change in number from 1990 to 2015, %	67.5%	68.2	71.8
Change in crude rate from 1990 to 2015, %	20.5%	21.0	23.6
Change in age-standardized rate from 1990 to 2015, %	–2.3	–3.0	–8.8
Deaths			
Number, 2015	1.2 million	2.4 million	1.2 million
Crude rate/100,000 population, 2015	16.7	32.9	16.2
Global rank, 2015 ^b	10th	11th	—
Change in number from 1990 to 2015, %	108.3	69.5	42.0
Change in crude rate from 1990 to 2015, %	49.9	21.9	2.2
Change in age-standardized rate from 1990 to 2015, %	14.3	–13.0	–29.6
Years of Life Lost Due to Premature Mortality			
Number, 2015	27.1 million	45.1 million	18.0 million
Crude rate/100,000 population, 2015	367.5	611.8	244.3
Global rank, 2015 ^b	17th	15th	—
Change in number from 1990 to 2015, %	62.1	49.1	33.1
Change in crude rate from 1990 to 2015, %	16.6	7.3	–4.2
Change in age-standardized rate from 1990 to 2015, %	1.2	–14.8	–30.2
Disability-Adjusted Life-Years⁷⁹			
Number, 2015	35.3 million	54.4 million	19.0 million
Crude rate/100,000 population, 2015	478.4	738.5	257.7
Global rank, 2015 ^b	20th	15th	—
Change in number from 1990 to 2015, %	63.3	52.1	34.7
Change in crude rate from 1990 to 2015, %	17.5	9.4	–3.1
Change in age-standardized rate from 1990 to 2015, %	0.4	–13.0	–29.3

^aLow GFR (glomerular filtration rate) as a risk factor resulted in the burden of CKD, cardiovascular disease (CVD), and gout.

^bGlobal rank for CKD stages 3–5 refers to a rank among 295 diseases and injuries; global rank for a low GFR refers to a rank among 79 risk factors expressed as a crude rate.

prevalent in females (195.4 million) than in males (127.1 million).

Almost one-third of subjects with a reduced estimated glomerular filtration rate (GFR; CKD stages 3–5) in 2015 lived in two countries: India (69.8 million, equal to 5.5% of the population) and China (47.8 million, 3.6% of the population). The United States, Brazil, and Japan each had more than 10 million people with a reduced GFR, and Russia, Germany, Pakistan, Indonesia, Bangladesh, Mexico, and Nigeria each had between 5 and 8 million. Two independent meta-analyses have also included albuminuria with a normal GFR in the definition of altered kidney function and estimated the burden of CKD to be 13.4% (with 10.6% attributed to stages 3–5) globally,⁷⁵ and 14.3% (with 9.8% attributed to stages 3–5) in geographically dispersed developing countries.¹⁵ It must be noted that these numbers do not fully take into account differences in age structure of the studied populations, which was included in the GBD study, and may explain the lower estimates of CKD prevalence in that study. Notwithstanding these differences, even the GBD study reported significant growth in the prevalence of CKD stages 3 to 5 since 1990—by 78.3% in absolute numbers and by 28.3% in crude rate. Moreover, there is significant heterogeneity among countries and regions, and the crude CKD prevalence rate is much higher in high-income countries (Fig. 75.4). In contrast, the age-standardized prevalence rate is higher among low- and middle-income countries (Fig. 75.5). The highest age-standardized prevalence of CKD stages 3 to 5 in 2015 was found in Southern and Western sub-Saharan Africa and South Asia (7285.5, 6462.4, and 6074.6/100,000 age-standardized population, respectively). Of concern, these rates were almost 50% higher than the mean global value. One of the prominent GBD study findings is that the change in age-standardized prevalence over 25

years, accounting for the dynamics of both age structure and population number, is much higher in the developing world (Fig. 75.6), whereas many HICs showed small increases or even a decline in age-standardized CKD rates. Importantly, the age-standardized prevalence rate increased significantly for 40 countries from 1990 to 2015, including India, Brazil, Pakistan, Mexico, Thailand, South Africa, and Iran, with each having more than 2 million people with CKD.

There is a significant disproportion between the crude and age-standardized rates of CKD, which is not always considered by policy makers. The crude prevalence rate (and thus the absolute number of patients requiring treatment) is higher in HICs (Fig. 75.7), mostly due to the longer life duration and higher proportion of older individuals. The age-standardized prevalence rate, which considers not only the size but also the age structure of a country's population, however, is much higher in LMICs (with a lower sociodemographic index; see Fig. 75.7), highlighting the fact that there are more young individuals of working age affected by CKD in lower-income compared with higher-income regions. This could suggest differences in causes, progression rates, and mortality of CKD by income region or lack of access to early detection and treatment of CKD in lower-income regions. It is well recognized that important differences in access to and quality of kidney care exist among different countries⁷⁶ and, even within the same country, different populations have higher risks of late referral to nephrology services.⁷⁷ Taken together, the higher frequency among younger individuals, limited access to kidney care, and inadequate access to essential medication contribute to the CKD burden, which has important implications for national economies, especially in countries with lower socioeconomic development.⁷⁸

It is estimated that in 2015, CKD resulted in 1.23 million deaths worldwide, being the direct cause of 1 in 43 global

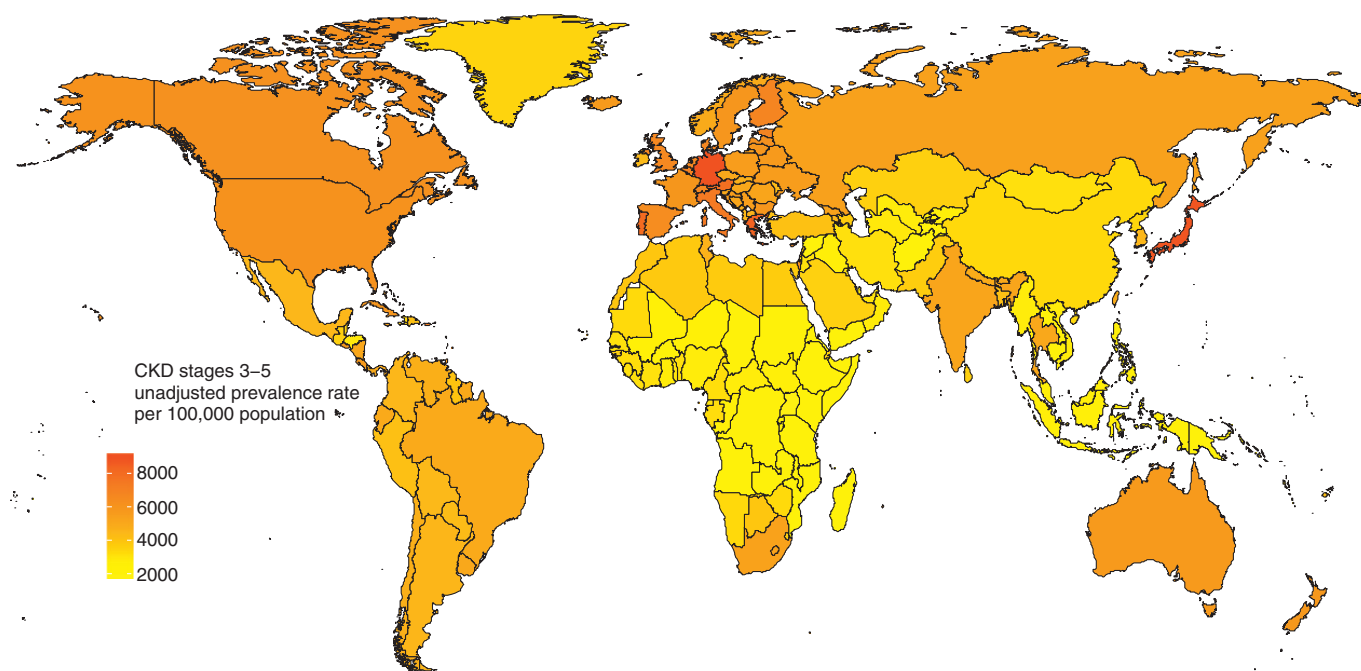


Fig. 75.4 World map of crude prevalence rate/100,000 population for chronic kidney disease (CKD) stages 3–5 in 2015 (both genders combined).

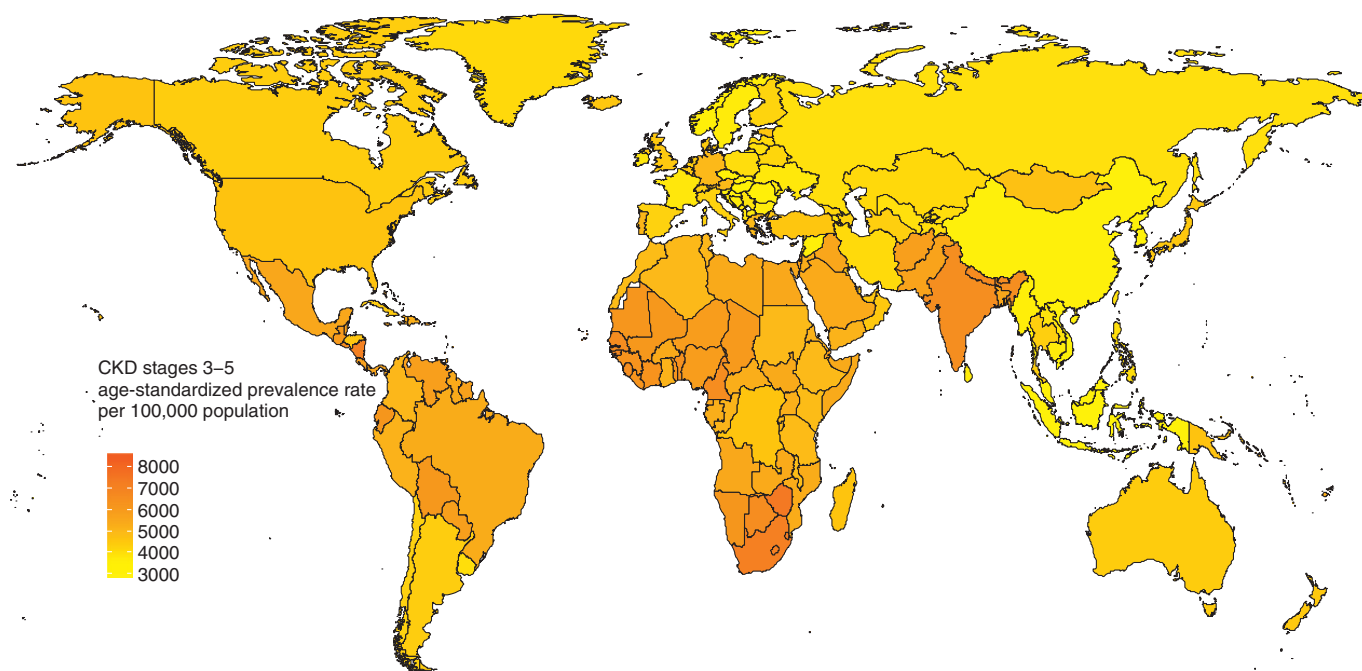


Fig. 75.5 World map of age-standardized prevalence rate/100,000 population for chronic kidney disease (CKD) stages 3–5 in 2015 (both genders combined).

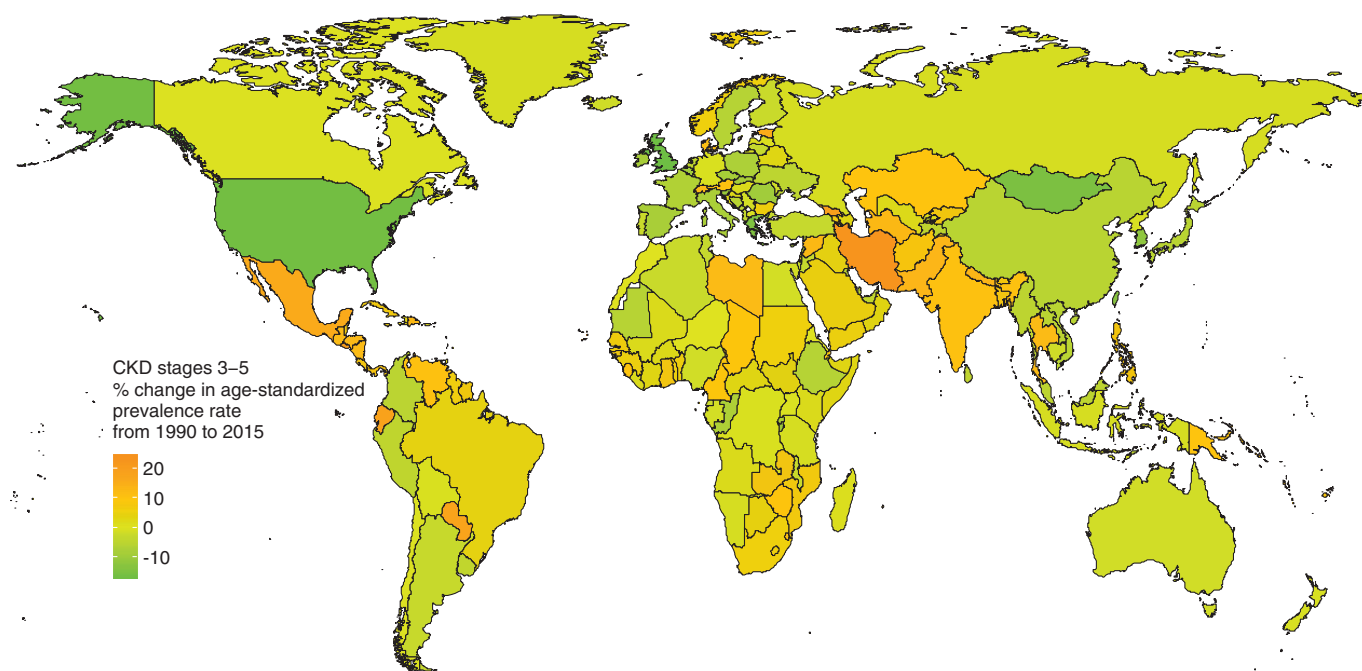


Fig. 75.6 World map of changes from 1990 to 2015 in age-standardized prevalence rate/100,000 population for chronic kidney disease (CKD) stages 3–5 (both genders combined).

fatal outcomes (see [Table 75.3](#)).² Of note, CKD is one of the few causes of death that is still increasing ([Fig. 75.8](#)). The global mortality rate from CKD has increased by 49.8% over the past 25 years, which far exceeds the percentage change of most other NCDs, infections, and injuries. This increase made CKD the 10th leading cause of death in 2015 among over 300 causes accounted for in the GBD study, compared with its ranking of 19th in 1990. Considering the burden

expressed in years of life lost (YLLs) due to premature death, the impact of CKD is somewhat lower, being the 17th leading cause of YLLs globally and accounting for 27.1 million YLLs in 2015. This observation reflects the fact that CKD affects more persons of advanced age. The aforementioned GBD study estimates of CKD mortality are based on sophisticated modeling that includes data on reported causes of death. A different approach to the estimation of CKD mortality, based

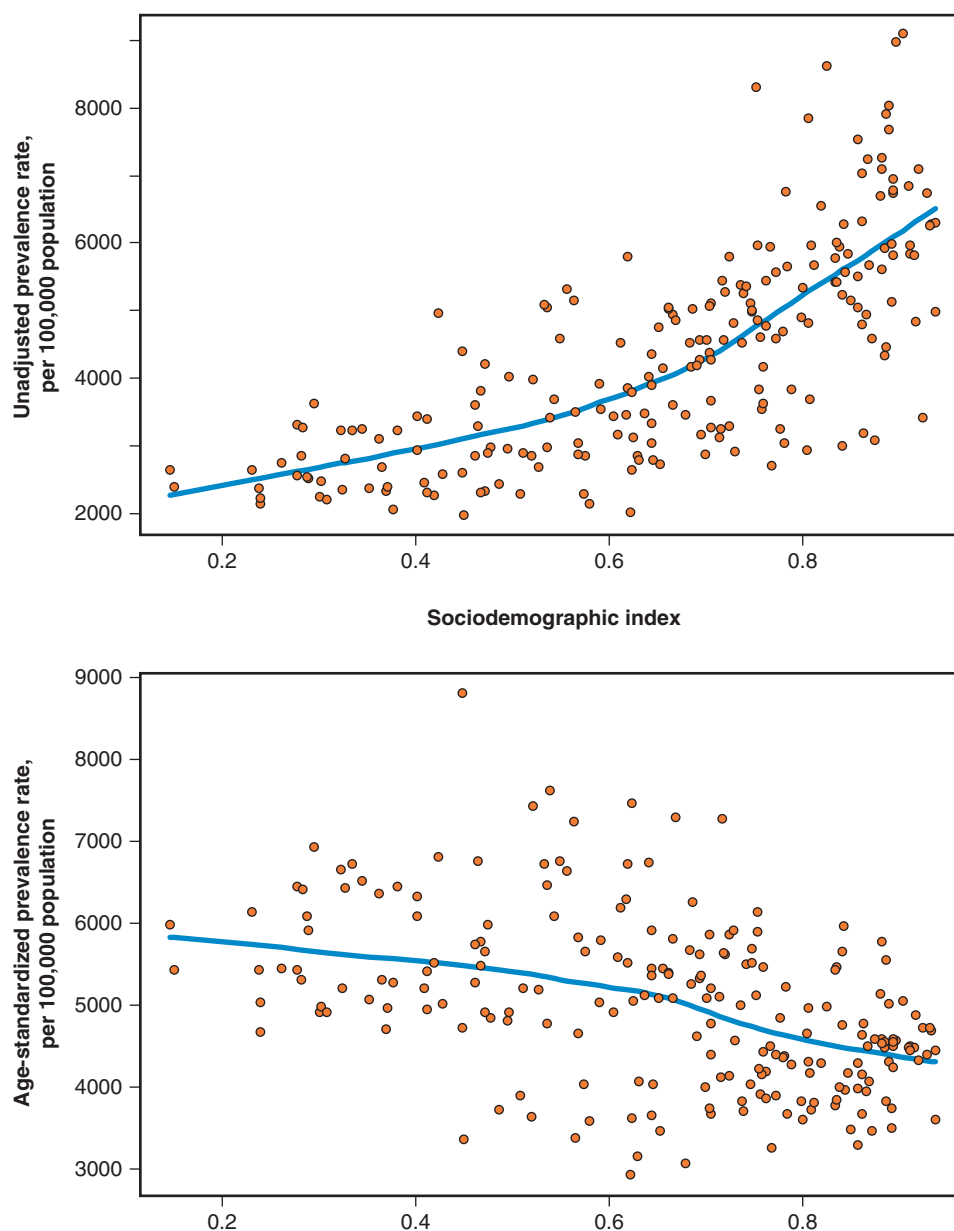


Fig. 75.7 Country-level correlation between sociodemographic index and chronic kidney disease (CKD) stages 3–5 prevalence rate/100,000 population in 2015 (both genders combined). *Upper panel* shows crude prevalence rate; *lower panel* shows age-standardized prevalence rate.

on the comparison of actual provision of RRT (dialysis and transplantation) and estimated RRT need,³ has suggested that between 2.3 and 7.1 million deaths worldwide could be attributable to death from ESKD due to lack of access to RRT. Lack of access to diagnosis of CKD in many lower-income regions hampers the accurate measurement of the true global CKD burden, which may therefore be underestimated in current studies.

Worldwide, the CKD-related number of years lived with disability (YLDs) accounted for almost 8.17 million in 2015, which corresponds to 1 of each 100 global YLDs from all diseases. This is an important number considering that the vast majority of individuals with CKD stage 3 are not included in the YLDs due to the GBD study methodology (see [Table 75.3](#)).⁷² Disability-adjusted life years (DALYs),⁷⁹ as a measure of disease burden, sums YLLs and YLDs, thus accounting for both mortality and

morbidity. CKD led to 35.26 million DALYs globally in 2015, reflecting a growth of 63.3% over 25 years. In contrast, the age-standardized DALY rate did not change significantly over the same period (0.4%), indicating that population growth and aging were the major drivers of global CKD DALYs.⁷⁹ However, age-standardized DALY rates did significantly increase in Oceania, Central Asia, high-income North America, and Central Latin America, highlighting a true increase in kidney disease burden, independent of demographic factors, and pointing to regions that require special attention to improve kidney health. Worldwide, the age-standardized DALY rate for CKD has increased from the 25th to the 20th position since 1990, in part because many other diseases demonstrated a decrease in this metric over the same period. All CKD burden indicators were substantially higher among those in older age groups ([Fig. 75.9](#)).

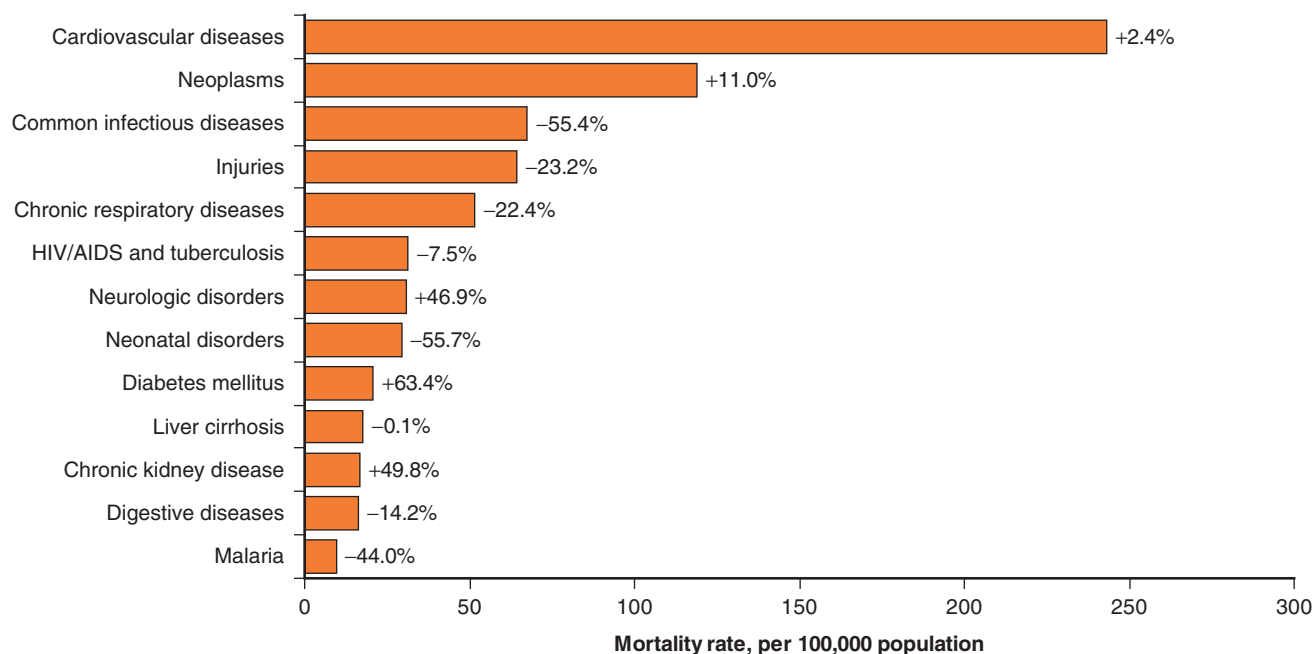


Fig. 75.8 Mortality crude rate for major classes of diseases and injuries. Bars represent mortality rate/100,000 population in 2015; numbers at the right side of bars represent percentage change in mortality rate between 1990 and 2015.

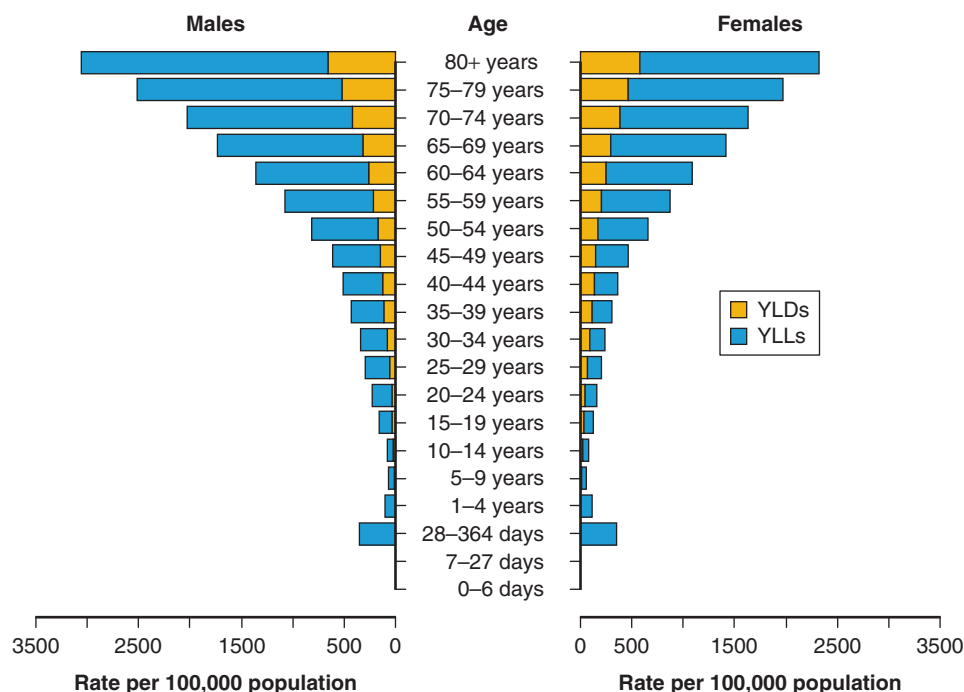


Fig. 75.9 YLL, YLD, and DALY rates/100,000 population for chronic kidney disease (CKD) stages 3-5 in 2015, for males and females according to age group. Orange bars represent YLDs, blue bars represent YLLs; total size of the bars equal to DALYs. DALYs, Disability-adjusted life years; YLDs, years lived with disability; YLLs, years of life lost.

MORTALITY AND MORBIDITY DUE TO CARDIOVASCULAR DISEASE DRIVEN BY LOW GLOMERULAR FILTRATION RATE

Importantly, kidney disease represents a direct cause of morbidity and mortality and is a risk factor for morbidity and mortality from other diseases. The GBD study has

recognized this double role of kidney disease and specifically considered a low GFR as a risk factor for CVD and gout.⁷⁴ The global number of YLDs attributed to low GFR includes those accounted for by CKD itself (8.17 million), as well as for CVD (0.98 million), and a minor fraction for gout (0.18 million). Thus, in total, a low GFR accounted for 9.34 million YLDs in 2015 (see Table 75.3). Worldwide, a low GFR resulted

in 27.1 million YLLs from CKD itself and 18.0 million YLLs for CVD, accounting for the total of 45.1 million YLLs in 2015. In other words, 1 of each 22 global YLLs, and 1 of each 18 YLLs from all risk factors in 2015 was attributed to a low GFR. A decreased GFR accounted for 1.19 million CVD deaths, which was almost equal to the 1.23 million deaths due to CKD itself. The number of CVD deaths attributed to a low GFR increased by 42%, whereas the number of deaths directly due to CKD stages 3 to 5 increased by 108% over 25 years. These numbers reflect less success in reducing mortality from kidney disease compared with CVD over this period. Within such growth, a low GFR became the 11th leading risk factor contributing to number of deaths in 2015 (ranked 17th in 1990). The total number of DALYs due to low GFR reached a total of 54.43 million in 2015, of which 35.26 million corresponded to CKD, 18.99 million to CVD, and 0.18 million to gout. Considering both absolute count and crude and age-standardized DALY rates/100,000 population, a low GFR ranked 15th among all 79 GBD risk factors considered, thus outranking such factors as low physical activity, drug use, unsafe sanitation, occupational injuries, and iron, vitamin A, and zinc deficiencies.⁷⁴ In total, one of each 19 DALYs in the world in 2015 was attributed to a low GFR.

The GBD study results permit expression of the burden of decreased GFR for each country and comparisons of the burden (expressed as DALY rates) of CKD and GFR-related CVD between countries (Fig. 75.10). In general, these two conditions are well correlated, but there are several countries with prominent discordance, which could be explained either

by the extremely high burden of CVD (as in former socialist countries) or an exceptionally high burden of CKD morbidity and mortality (as in Mexico, El Salvador, and Thailand). On the global scale, CVD accounted for 11.3% of all age-standardized YLLs and 36.5% of all age-standardized DALYs due to a low GFR. From the cardiovascular perspective, a low GFR was responsible for 4.0% of all age-standardized YLLs and 5.7% of all age-standardized DALYs for CVD in 2015. In other words, a low GFR resulted in one of each 25 CVD age-standardized YLLs and one of each 17 CVD age-standardized DALYs. This was almost double the impact on CVD from excessive alcohol use, comparable with the impact of low physical activity, and only three times lower than the impact of high fasting plasma glucose levels or smoking as CVD risk factors.⁷⁴

SCOPE OF AMENABLE CHRONIC KIDNEY DISEASE MORTALITY AMENABLE TO INTERVENTION

In LLMICs, the burden of CKD is much higher as compared with high-income regions, especially if age-standardized rates are considered. The major reason for the higher burden in LLMICs is the excessive mortality due to lack of access to RRT and higher rates of fatal CVD events where health facilities are inadequate or universal health coverage is not in place. In countries where CKD-related mortality is higher, a substantial part of the population would not survive to live with disability; therefore, comparison of the ratio of YLLs to YLDs as composite indicators of mortality and morbidity would provide important information that could be used as a

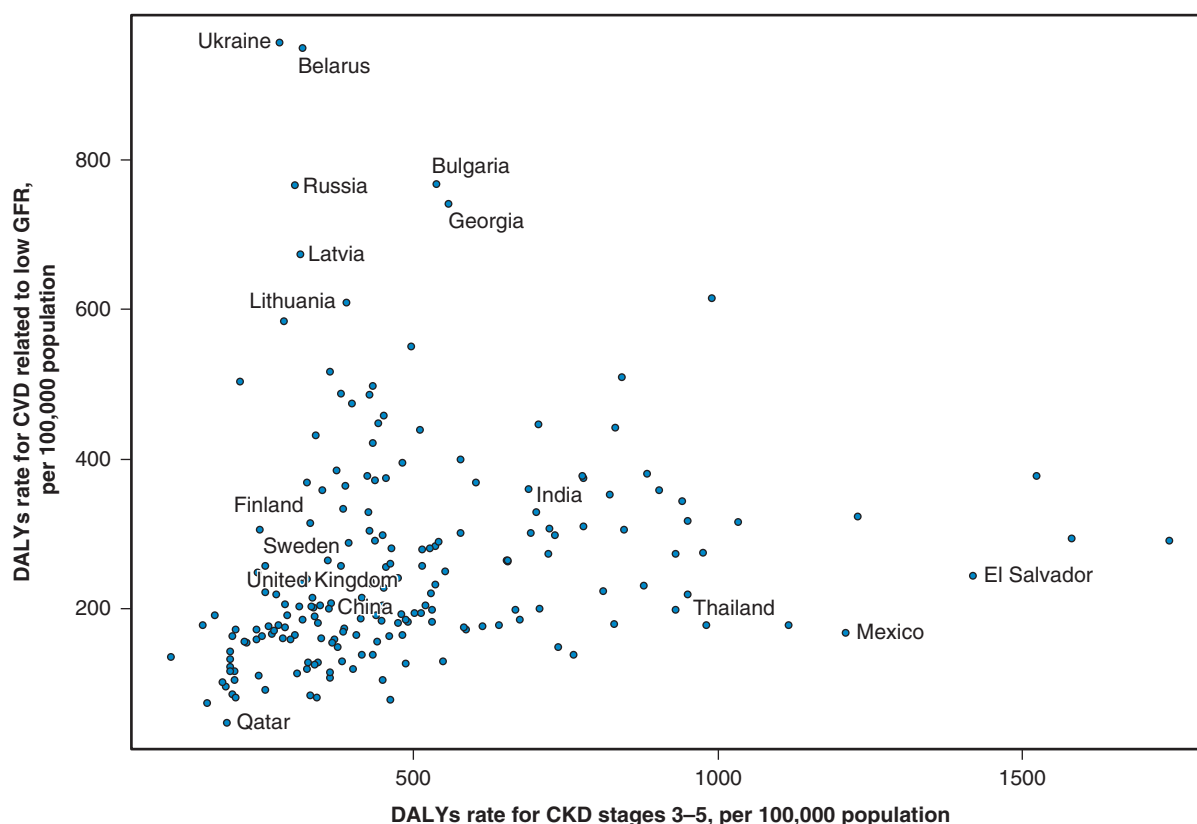


Fig. 75.10 Country-level DALY crude rates/100,000 population for chronic kidney disease (CKD) stages 3–5 and cardiovascular diseases (CVD) attributed to low glomerular filtration rate (GFR) in 2015 (both genders combined). DALY, Disability-adjusted life-year.

surrogate marker of amenable mortality (Fig. 75.11). In HICs, the ratio of YLDs to YLLs for CKD is close to 1 (e.g., CKD in Western Europe accounted for 676.7 thousand YLDs and for 822.7 thousand YLLs in 2015). This reflects the universal provision of RRT that prevents a substantial proportion of immediate CKD deaths and the longer duration of a healthy life, which reduces both morbidity and mortality. An exception to this model, however, is the United States where, despite being the world's largest economy, the number of YLLs is almost twice as high as YLDs, indicating a greater impact of CKD mortality. This observation highlights the importance of universal health coverage because its lack leads to late referral to nephrology services and higher rates of complications and mortality.⁸⁰

A more critical situation exists in regions with both a shortage of RRT treatment and mainly OOP health expenditures,

where YLLs prominently predominate over YLDs; the ratio is close to 5 in Southern sub-Saharan Africa and to 15 in Oceania. In such emerging regions, data are sparse but, as discussed earlier, even after the start of dialysis, a vast majority of patients are forced to withdraw from this life-saving treatment (up to 85% of the incident patients) and die because of their inability to pay for RRT.^{3,35} The difference between YLLs and YLDs illustrates how many lives could be saved by implementing appropriate prevention strategies, conducting early kidney disease detection programs, educating health personnel, improving access to nephroprotective treatment and RRT, and preventing CKD and CVD complications. Considering the global numbers of 27.1 million YLLs and 8.2 million YLDs due to CKD in 2015, the reduction of amenable CKD mortality in 2015 would have been expected to save almost 20 million YLLs globally.

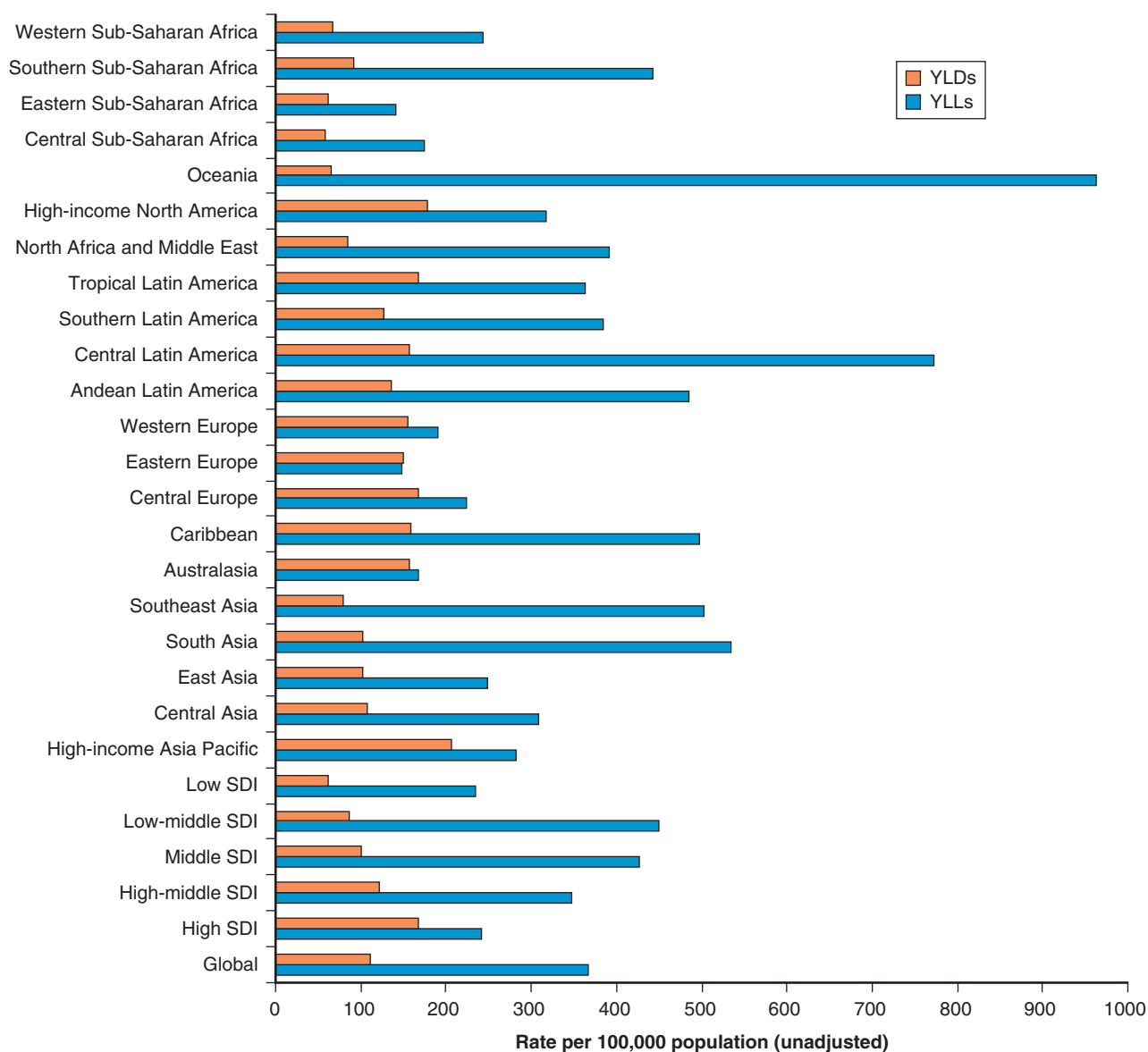


Fig. 75.11 Global and regional burden of chronic kidney disease (CKD) stages 3–5 in YLLs and YLDs crude rates/100,000 population for 2015 (both genders combined). YLDs, Years lived with disability; YLLs, years of life lost.

DRIVERS OF EPIDEMIOLOGIC TRANSITION FOR CHRONIC KIDNEY DISEASE

The changes in epidemiology of CKD over recent decades may have various causative factors. First, CKD morbidity and mortality are related to the global processes of population growth, aging, and migration. The world's population grew from 5.3 billion in 1990 to 7.3 billion in 2015. In the same period, the median global age increased from 24 to 30 years, and the proportion of urban populations increased from 43% to 53.8% (<http://www.un.org/en/development/desa/population/theme/trends/index.shtml>). These changes directly affect the epidemiology of CKD, with absolute counts of CKD patients increasing in proportion to population growth and aging and possibly also to improved health outcomes in the urban populations. This epidemiologic transition also drives the metabolic and nutritional risk factors for CKD development and progression. Thus, over 25 years, the CKD age-standardized DALY rates attributable to arterial hypertension have increased by 6.8%, to high fasting plasma glucose levels by 15.0%, to high body mass index by 34.2%, and to high-sodium diet by 5.6%. During the same period, however, the DALY rates attributable to these risk factors have decreased in the case of CVD.⁷⁴

A second important driver of CKD growth, especially relevant to the increase in LMICs, is the persistent inequality among different population subgroups in exposure to risk factors, as well as access to prevention and treatment. Usually countries with lower economic development have higher rates of CKD and of NCDs in general.⁸¹ However, even within the same country, substantial inequalities exist in access to medical care and RRT in particular.⁸² Thus, in many regions, medical care for rural populations is insufficiently developed and available, and one would expect a higher mortality from CKD and ESKD, even if the prevalence of CKD risk factors might be lower in rural settings. CKD is also more prevalent among the urban poor than those of higher socioeconomic status in developed countries.⁸³ Other vulnerable groups include women, older persons, and children who in many societies still have reduced access to medical treatment. Women have a higher CKD prevalence but generally a lower rate of CKD progression as compared with men. In addition, there is evidence of lower access to kidney treatment for women.⁸⁴ Moreover, CKD in pregnant woman substantially increases the risk of preeclampsia, preterm delivery, small-for-gestational-age and low-birth-weight infants, all factors associated with low nephron number and increased risks of CKD and other NCDs over the life course (see Chapter 21).^{85,86} Thus, persisting inequalities and lack of attention to women's health form a vicious circle that contributes to the CKD burden over generations. It could be speculated that current higher age-standardized rates of CKD burden in LLMICs may in part be affected by suboptimal fetal and early childhood nutrition and development leading to CKD in later life. Future generations may see the impact of the current humanitarian crises on fetal health and CKD epidemiology scattered worldwide over the coming decades.

A third reason underlying the CKD epidemiologic transition is the increase in CKD burden in certain countries and whole regions, independent of changes in the general population. Positive and significant dynamics of age-standardized CKD rates across regions highlight the most problematic countries

where investigation is urgently needed to examine and address the factors leading to increasing age-standardized rates. An example of such an emerging CKD epidemic is “CKD of unknown cause,” CKDu), which affects predominantly adult agricultural workers in subequatorial countries of Central America and South Asia. Available data have not permitted an estimate of the burden of CKDu over time, but in some regions such as Nicaragua, CKD is the primary cause of death, fueled by the epidemic of CKDu.² Additional evidence, not well represented in the scientific literature, has suggested that other groups of workers (e.g., metal and diamond miners, industrial paint shop workers) with high exposure to environmental or occupational hazards are also affected by nephropathies that require further investigation and efforts at prevention.^{87–89} It is possible that worldwide, millions of people work or live in unsafe environments, which adds to the global CKD burden.

A fourth factor influencing CKD estimates is the precision of estimated GFR (eGFR) equations (i.e., the correspondence between measured and estimated GFR) and the availability of repetitive testing to confirm chronic persistence of abnormal kidney disease markers. Use of different GFR estimation equations may lead up to a 30% difference in CKD prevalence estimates. A meta-analysis analyzing data from 1.1 million individuals has found that the proportion of subjects with a low GFR decreased from 8.7% or 6.3% with the use of the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation relative to the Modification of Diet in Renal Disease Study (MDRD) equation.⁹⁰ Because a substantial amount of data on CKD prevalence in the GBD study was obtained by using the MDRD equation, the true CKD estimates in the studied populations could be substantially lower. Of note, in the GBD study, the reference equation was MDRD until 2015, but was switched to the CKD-EPI equation as of 2016 with the application of a specific crosswalk procedure. Consequently, a decrease in CKD prevalence estimates will occur in the GBD study in comparison with the aforementioned numbers simply because of the use of a different reference eGFR equation. These observations demonstrate the importance of a good instrument to measure kidney function for both individual patient and population studies to best inform planning of the necessary health interventions.

ACUTE KIDNEY INJURY AND THE OBY25 INITIATIVE

The Oby25 initiative of the International Society of Nephrology (ISN), launched in 2013, aimed to raise awareness about the preventability and treatability of AKI worldwide, targeting no preventable deaths from AKI by 2025. As described earlier, AKI is associated with significant morbidity and mortality, but data have not thus far been reported by the GBD study. One aim of the Oby25 initiative was to determine the burden of AKI. Two large meta-analyses and systematic reviews were performed, which reported that AKI affected 21% of hospital admissions worldwide.^{4,19,24} Most cases of AKI were Kidney Disease: Improving Global Outcomes (KDIGO) stage 1, and dialysis was required in 11% of cases.⁴ Pooled mortality was 21%. The Oby25 investigators also conducted a global snapshot of AKI, inviting clinicians to enter data about patients with AKI over a 2-month period.²⁴ The most common causes of AKI were hypotension in HICs and dehydration in LLMICs. Dialysis requirement and mortality were higher in LMICs, likely reflecting the severity of illness (lateness)

on presentation. The meta-analyses and the global snapshot are, however, inherently biased by publication availability and by elective participation in the snapshot; therefore, the true global picture of AKI remains unknown. What is clear is that the causes of AKI differ across the globe, depending on prevalence rates of diabetes and hypertension, infections, and public health interventions.^{20,22,91} As with CKD, AKI in LLMICs tends to affect younger individuals and is more severe on presentation to hospital.^{18,19,21,22} The 0by25 initiative has, however, served to raise awareness about AKI globally. Future iterations of the GBD study will attempt to report AKI data better, which will be an important step in achieving a more complete picture of the global burden of kidney disease.

SAVING YOUNG LIVES INITIATIVE

There has been a global push for universal access to dialysis for AKI through the 0by25 and Saving Young Lives (SYL), given that dialysis need should be temporary because renal function can recover and therefore is less expensive per patient than lifelong treatment for ESKD.⁴ The SYL initiative, launched as a combined endeavor between the International Pediatric Nephrology Association, ISN, International Society of Peritoneal Dialysis, and European Peritoneal Dialysis in 2012, has saved the lives of several hundred individuals (mostly children) over the past 5 years and has trained and mentored local physicians and nurses to perform acute PD⁹² using home-made dialysis solutions and innovative use of alternative catheters (Table 75.4).^{93,94} This program has remained dependent on external funding and is highly reliant on individual champions. Unexpected challenges, outlined by those implementing SYL programs, include lack of access to routine laboratory testing, staff changes, unreliability of financing sources, cultural differences, and lack of referrals to SYL centers.⁹⁴ These challenges highlight the crucial importance of program integration into existing health care system structures, which is important to avoid generation of

isolated programs and contribute to strengthening of the health care system.

THE GLOBAL KIDNEY HEALTH ATLAS

The Global Kidney Health Atlas (GKHA), as part of the “Closing the Gaps” initiative of the ISN, has provided an overview of the current state of kidney care and how it is organized and structured across the world, focusing on existing kidney care (AKI, CKD), awareness levels (based on existence of relevant policies and guidelines), care structures, and workforce for optimal care delivery.^{11,95,96} A total of 125 United Nations Member States (encompassing 93% of the world’s population) responded to the survey, as described in more detail elsewhere.^{97–102} For comprehensiveness, the elements of the survey were evaluated within the framework of the health systems building blocks outlined by the WHO—leadership and governance, health care financing, health care workforce, medical products and technologies, information and research, and service delivery.¹⁰³ Study respondents across countries included national nephrology society leaders, policy makers, and patient organization representatives identified by the country and regional nephrology leadership through the ISN.⁹⁵ As expected, there was significant variation across countries on the key survey elements discussed later.^{97–102}

AWARENESS OF KIDNEY HEALTH AS A GOVERNMENT PRIORITY, STRATEGIES, GUIDELINES, AND ADVOCACY

CKD was recognized as a health care priority (defined as outlining principles and/or defining practices) by government in 36% of countries overall ($n = 42/116$) and in 53% ($n = 9/17$), 52% ($n = 16/31$), 20% ($n = 6/30$), and 29% ($n = 11/38$) of low, lower middle, upper middle, and high-income countries, respectively. Overall, 42% of countries ($n = 49/116$) reported an advocacy group at higher levels of government to raise the profile of CKD and its prevention, whereas only

Table 75.4 Preparation of Dialysis Solution Using Commercially Available Intravenous Fluid^a

Type of Fluid	Na ⁺	K ⁺	Ca ²⁺	Mg	Cl ⁻	HCO ₃ ⁻	Lactate	pH	Osmolarity
Hartmann’s solution	131	5	2.0		111		29	7.0	278
Ringer’s lactate	131	5	1.8		112		28	6.5	279
Plasmalyte B	130	4	0	1.5	110	27		7.4	273
Half-normal saline	77				77			5.0	154

^aPreparation of dialysis solutions using the above intravenous solutions:

- 1 L Plasmalyte + 30 mL 50% dextrose (15 g) will generate a solution with the following concentrations: glucose, 1.45%; Na, 126 mmol/L; HCO₃⁻, 27 mmol/L; K, 3.8 mmol/L; Mg, 1.45 mmol/L; osmolarity = 342 osm.

This is very similar to some of the bicarbonate-based solutions sold by industry:

- 1 L Ringer’s lactate + 30 mL 50% dextrose (15 g) will generate a solution with the following concentrations: Na, 127 mmol/L; lactate, 27 mmol/L; Ca, 1.36 mmol/L; K, 3.8 mmol/L; glucose 1.45 %; osmolarity = 346 osm.

This is similar to lactate-based peritoneal dialysis solutions.

NOTE: The above solutions both contain potassium.

- 1 L half-normal saline + 40 mL 8.5% NaHCO₃ (40 mmol) + 40 mL 50% dextrose (20 g) + 60 mL 3% NaCl (30 mmol) will generate a solution with approximately the following concentrations: Na, ± 130 mmol/L; HCO₃⁻, 35 mmol/L; glucose, 1.7%; osmolarity = 340 osm.

Na⁺, Sodium; K⁺, potassium; Ca²⁺, calcium; Mg, magnesium; Cl⁻, chlorine; HCO₃⁻, bicarbonate.

Adapted from Cullis Abdelraheem M, Abrahams G, et al. Peritoneal dialysis for acute kidney injury. *Perit Dial Int*. 2014;34(5):494–517.

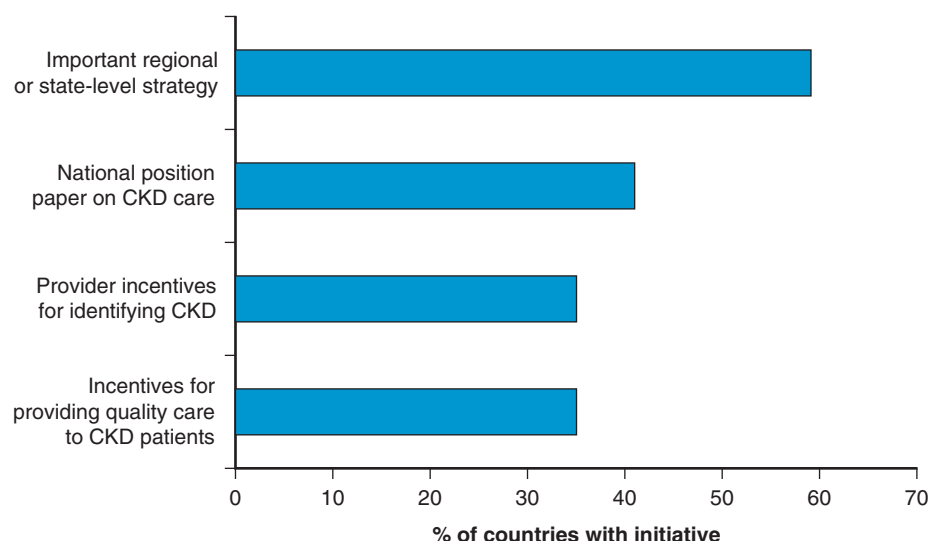


Fig. 75.12 Reported distribution of initiatives for promoting chronic kidney disease (CKD) as a health care priority among countries with no national strategy for CKD. Of the 43 countries with no CKD strategy, 34 (79%) responded to this question. Percentages do not total 100% because countries were able to select more than one option. (Data from Global Kidney Health Atlas. https://www.theisn.org/images/ISN_Biennial_Report_2011-2013/GKHAAtlas_Linked_Compressed1.pdf.)

19% of countries ($n = 21/116$) reported advocacy for AKI.^{95,99} A lack of government priority for kidney health could represent a lack of awareness of CKD or the prioritization of other political issues.

It is also necessary to consider that the definition of health priority differs across countries. For example, a priority in some regions could represent a focus on prevention, whereas in others it could refer to increasing access to treatment. Advocacy for AKI was less than half that for CKD. The reduced attention given to AKI has been recognized, and the 0by25 initiative is one response to this. Dissemination of this strategy aims to increase advocacy and awareness of the importance of preventing AKI. Both CKD and AKI advocacy groups were more common in LICs compared with HICs; however, details regarding the roles or actions of the advocacy groups were not reported.

Overall, 17% of countries ($n = 19/109$) reported a national strategy for improving awareness and optimal care for non-dialysis CKD patients, 43% for chronic dialysis patients ($n = 45/104$), and 40% for kidney transplant patients ($n = 40/100$). Conversely, 37% of countries ($n = 43/116$) reported no national strategy for CKD care (irrespective of nondialysis CKD, chronic dialysis, or transplantation). These countries reported other initiatives that identified CKD as a health care priority, including strategies at a regional or state level, a national position paper on CKD (a document providing an overview of information and recommendations for kidney care, but not mandated by legislation), and incentives for identifying CKD and providing quality care to CKD patients (Fig. 75.12). For AKI, 49% of countries ($n = 57/116$) reported at least one strategy for improving the identification of AKI (Fig. 75.13). The most common strategies were having tools available for the detection and management of AKI (32%; $n = 37/116$) and increasing access to acute dialysis facilities (31%; $n = 36/116$); 16% of countries ($n = 19/116$) reported having a national position paper on AKI identification and care.

Overall, national strategies for enhancing awareness of kidney disease were quite low, particularly in LICs. Because of their importance in providing consistent high-quality and safe care and for standardization of metrics for evaluating quality and outcomes of care, the development of national strategies to address kidney disease should receive more attention. Professional guidelines and care frameworks are vital for enhancing the understanding of AKI and CKD care. Despite the significant work of the ISN and a related organization (KDIGO) to facilitate adoption of practice guidelines in kidney care across countries, this is not yet universal. Access to management and referral guidelines was more common for CKD (79%; $n = 92/116$) than AKI (53%; $n = 61/116$; Table 75.5). A lack of guidelines was more common among LICs compared with HICs for both AKI and CKD. AKI guidelines focused on the identification of AKI in inpatient (95%; $n = 57/60$) and outpatient (67%; $n = 40/60$) settings, access to dialysis treatment (93%; $n = 56/60$), timing and urgency for nephrology referral (80%; $n = 48/60$), and protocols for mitigating risk of AKI in specific situations (70%; $n = 42/60$). CKD guidelines covered complication management (86%; $n = 79/92$), identification of CKD progression (88%; $n = 81/92$), timing and urgency of nephrology referral (87%; $n = 80/92$), risk factor management (84%; $n = 77/92$), and multidisciplinary care approaches (71%; $n = 65/92$).

Of note, in 50% of countries rated, both awareness and adoption of CKD guidelines among non-nephrologist physicians was suboptimal. This may reflect an equally low level of awareness about CKD in general. Awareness of CKD among non-nephrologist specialists was reported as low or below average (based on perception of survey respondents of what they deemed to be level of awareness about CKD guidelines in their countries) by 49% of countries and as average by 35% of countries and, among primary care physicians, as low or below average by 50% of countries. With the recognition that non-nephrologist physicians have low awareness and

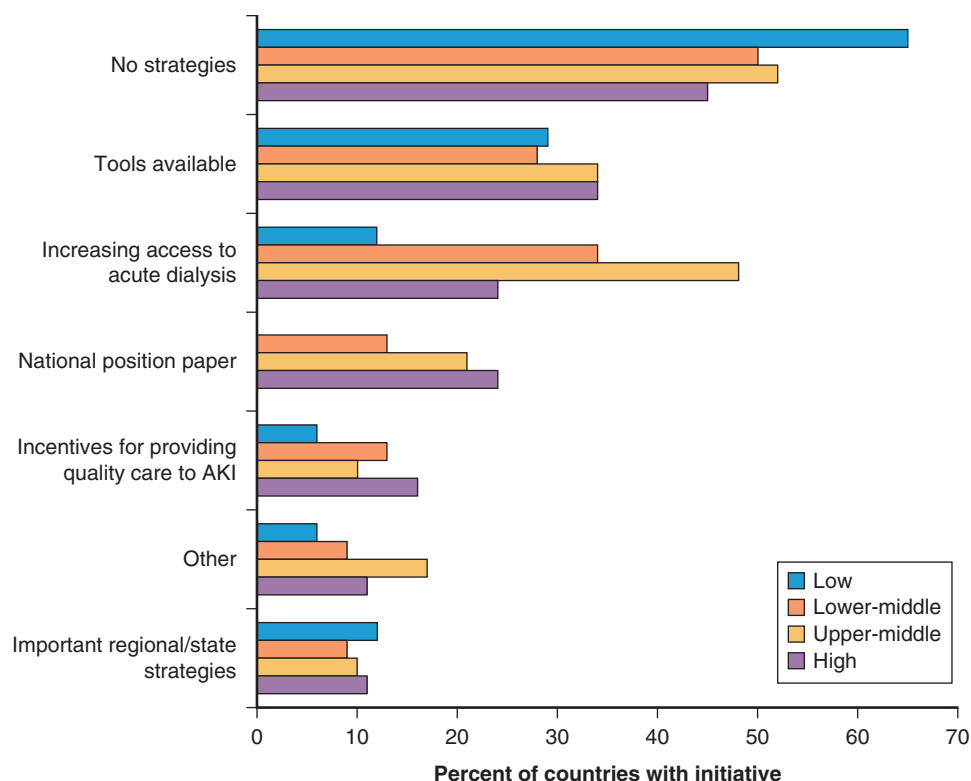


Fig. 75.13 National policies and strategies for identifying acute kidney injury (AKI) by World Bank income group across 116 countries. Percentages do not total 100% because countries were able to select more than one option. (Data from Global Kidney Health Atlas. https://www.theisn.org/images/ISN_Biennial_Report_2011-2013/GKHAAtlas_Linked_Compressed1.pdf.)

Table 75.5 Availability of and Access to Management and Referral Guidelines for Kidney Care Across 116 Countries

Region	AKI Management and Referral Guidelines				CKD Management and Referral Guidelines			
	National N (%)	Regional N (%)	International N (%)	None N (%)	National N (%)	Regional N (%)	International N (%)	None N (%)
Overall	8 (7)	1 (1)	52 (45)	55 (47)	31 (27)	1 (1)	60 (52)	24 (21)
ISN Region								
Africa	2 (7)	0 (0)	8 (27)	20 (67)	4 (13)	0 (0)	13 (43)	13 (43)
Eastern and Central Europe	0 (0)	0 (0)	15 (94)	1 (6)	2 (13)	0 (0)	13 (81)	1 (6)
LAC	1 (6)	0 (0)	7 (44)	8 (50)	9 (56)	0 (0)	7 (44)	0 (0)
Middle East	0 (0)	0 (0)	5 (38)	8 (62)	1 (8)	1 (8)	8 (62)	3 (23)
NIS and Russia	2 (33)	0 (0)	3 (50)	1 (17)	1 (17)	0 (0)	5 (83)	0 (0)
North America	0 (0)	0 (0)	2 (100)	0 (0)	1 (50)	0 (0)	1 (50)	0 (0)
North and East Asia	0 (0)	0 (0)	2 (33)	4 (67)	3 (50)	0 (0)	3 (50)	0 (0)
OSEA	2 (15)	1 (8)	3 (23)	7 (54)	6 (46)	0 (0)	2 (15)	5 (38)
South Asia	0 (0)	0 (0)	1 (20)	4 (80)	0 (0)	0 (0)	3 (60)	2 (40)
Western Europe	1 (11)	0 (0)	6 (67)	2 (22)	4 (44)	0 (0)	5 (56)	0 (0)
World Bank Income Group								
Low Income	0 (0)	0 (0)	3 (18)	14 (82)	1 (6)	0 (0)	6 (35)	10 (59)
Lower middle income	5 (16)	0 (0)	11 (35)	15 (48)	7 (23)	0 (0)	16 (52)	8 (26)
Upper middle income	1 (3)	1 (3)	14 (47)	14 (47)	9 (30)	0 (0)	16 (53)	5 (17)
High Income	2 (5)	0 (0)	24 (63)	12 (32)	14 (37)	1 (3)	22 (58)	1 (3)

AKI, Acute kidney injury; CKD, chronic kidney disease; ISN, International Society of Nephrology; LAC, Latin America and the Caribbean; NIS, newly independent states (of the former Soviet Union); OSEA, Oceania and Southeast Asia.

Adapted from GBD 2015 Mortality and Causes of Death Collaborators. Global, regional, and national life expectancy, all-cause mortality, and cause-specific mortality for 249 causes of death, 1980–2015: a systematic analysis for the Global Burden of Disease Study 2015. *Lancet*. 2016;388(10053):1459–1544; and from Global Kidney Health Atlas. https://www.theisn.org/images/ISN_Biennial_Report_2011-2013/GKHAAtlas_Linked_Compressed1.pdf.

adoption of CKD guidelines, it is important to determine if it is due to lack of distribution, accessibility, or interest. Even in cases where national or regional guidelines do not exist, international guidelines should be accessible; however, if barriers such as Internet access or language prevent the distribution or adoption of guidelines, these issues should be addressed. Similar to CKD guidelines, nearly 50% of countries rated both awareness and adoption of AKI guidelines among non-nephrologist physicians as low or below average, and this may correspond with the limited awareness of AKI in general among non-nephrologist specialists and primary care physicians. Notably, the level of awareness and level of adoption of AKI guidelines were similar, suggesting that awareness may not be as much of an issue as reasons why guidelines are not being adopted. Further efforts to increase awareness of AKI and increase adoption of guidelines are warranted.

INFRASTRUCTURE FOR TREATMENT OF ACUTE KIDNEY INJURY AND CHRONIC KIDNEY DISEASE

The GKHA was the first initiative to assess global capacity for kidney care in terms of the key building blocks of a functional health system and to evaluate the readiness of countries and regions to enhance such care.^{95,98,99,101} There were significant gaps in services, workforce, and capacity for research in some countries and regions. For example, most countries in Africa had no facilities for PD or kidney transplantation. Few countries had public funding for RRT services and medications for CKD care (including dialysis and transplant); there was a large private contribution toward payment for RRT services and medications, particularly in countries across Africa, South Asia, Oceania, and Southeast Asia.⁹⁷ More than two-thirds of countries reported the absence of capacity to participate in clinical research.¹⁰⁰

Availability of Chronic Kidney Disease Monitoring and Management Services

There has been wide variation in terms of availability of services for CKD monitoring and management at the primary, secondary, and tertiary care levels across countries based on income groups (Fig. 75.14).^{95,104} LMICs, especially in Africa, had limited services for the diagnosis, management, and monitoring of CKD at the primary care level, with only 12% having serum creatinine level measurement, including eGFR, available. In LICs, 29% had access to qualitative urinalysis using urine test strips; however, no LICs had access to determining the urine albumin-to-creatinine ratio (UACR) or urine protein-to-creatinine ratio (UPCR) measurements at the primary care level (see Fig. 75.14A). Across all income groups, availability of services in the secondary/tertiary care level was considerably higher compared with the primary care level (see Fig. 75.14B).

Availability of Renal Replacement Therapy Services

Dialysis. All countries reported having chronic HD services, and more than 90% of countries reported having acute HD (see Fig. 75.14C).^{95,98} More than 90% of upper middle and high-income countries reported having chronic PD services, with 64% and 35% availability in LMICs and LICs, respectively. In comparison, acute PD had the lowest availability across all income groups (see Fig. 75.14B).

Kidney Transplantation. More than 90% of upper middle and high-income countries reported having kidney transplantation services, with more than 85% of these countries reporting the organ source as a combination of living and/or deceased donors (see Fig. 75.14D). As expected, LICs had the lowest availability of kidney transplantation services, with only 12% reporting availability and live donors as the only source.

Workforce for Care Delivery

Considerable variation was noted in the density of nephrologists across countries. The lowest density (<5 nephrologists/million population) was very common in LICs, whereas the highest density (>15 nephrologists/million population) was mainly reported in HICs (Fig. 75.15).^{95,101} Most countries reported nephrologists as primarily responsible for both CKD and AKI care. Primary care physicians (PCPs) had more responsibility for CKD care than AKI care; 64% of countries reported that PCPs were primarily responsible for CKD and 35% for AKI. Intensive care specialists were primarily responsible for AKI in 75% of countries; this is understandable because AKI is an acute condition typically treated in the hospital. However, only 45% of LICs reported that intensive care specialists were primarily responsible for AKI, compared with 90% of HICs. This discrepancy may be due to a general shortage of intensive care specialists in LIC. The appropriate number of nephrologists in a country is dependent on many factors, including need, priority, and resources and, as such, there is no global standard with respect to nephrologist density. Nevertheless, the density in LICs suggests a shortage of nephrologists, which is problematic because nephrologists are essential to provide leadership, and a lack of nephrologists may result in consequences for policy and practice. Encouragingly, the number of nephrologists and nephropathologists is rising in LICs, in part due to the existence of fellowship programs supported by international nephrology organizations.¹⁰⁵ Importantly, the role of a nephrologist may differ, depending on how the health care system is structured. In some regions, kidney disease patients are primarily cared for by PCPs, so the need for nephrologists may differ compared with other regions that primarily depend on a nephrologist for managing kidney disease patients.

The density statistics merely represent the number of nephrologists per million population and provide no indication of the adequacy to meet the needs of the population or quality of care, which depend on the volume of patients with kidney disease and workforce support. For the other care providers essential for kidney care, there were variations with distribution (availability and adequacy) across countries. Overall, provider shortages were highest for renal pathologists (86% of countries reported a shortage), vascular access coordinators (81%), and dietitians (78%), and shortages were more common in LICs. A minority of countries (35%) reported a shortage in laboratory technicians (Fig. 75.16).

Capacity for Kidney Research

There were significant variations across countries regarding the infrastructure available for optimal conduct of kidney research, including availability of funding agencies and structures, capacity building and training systems, regulatory frameworks, and storage facilities.^{95,100} Only 27% of countries reported a national agency for funding clinical trials, and

these were more common in HICs (45%) compared with LICs (12%). Overall, 15% of countries did not participate in clinical trials on kidney disease, and LICs had lower participation across all phases (1–4) of clinical trials. Moreover, less than half of countries (46%) had formal training for physicians in clinical trial conduct, and even fewer countries (34%) had formal training for nonphysicians or research assistants and associates involved in clinical trials. Although 85% of countries had the capacity (trained workforce) to conduct observational cohort studies, fewer countries (48%)

had funding to conduct the studies, particularly LICs. Nonetheless, 91% of countries had ethics approval for observational studies in CKD, 62% of which were managed by an institutional regulatory agency. Nearly half of countries (47%) had academic centers for coordinating and monitoring sites for renal clinical studies; however, this was more common in HICs (63%) compared with LICs (12%).

Similarly, the capacity for storing clinical trial medications was low across LICs. Storing medication requires equipment, electricity, facilities, and other resources. It has been

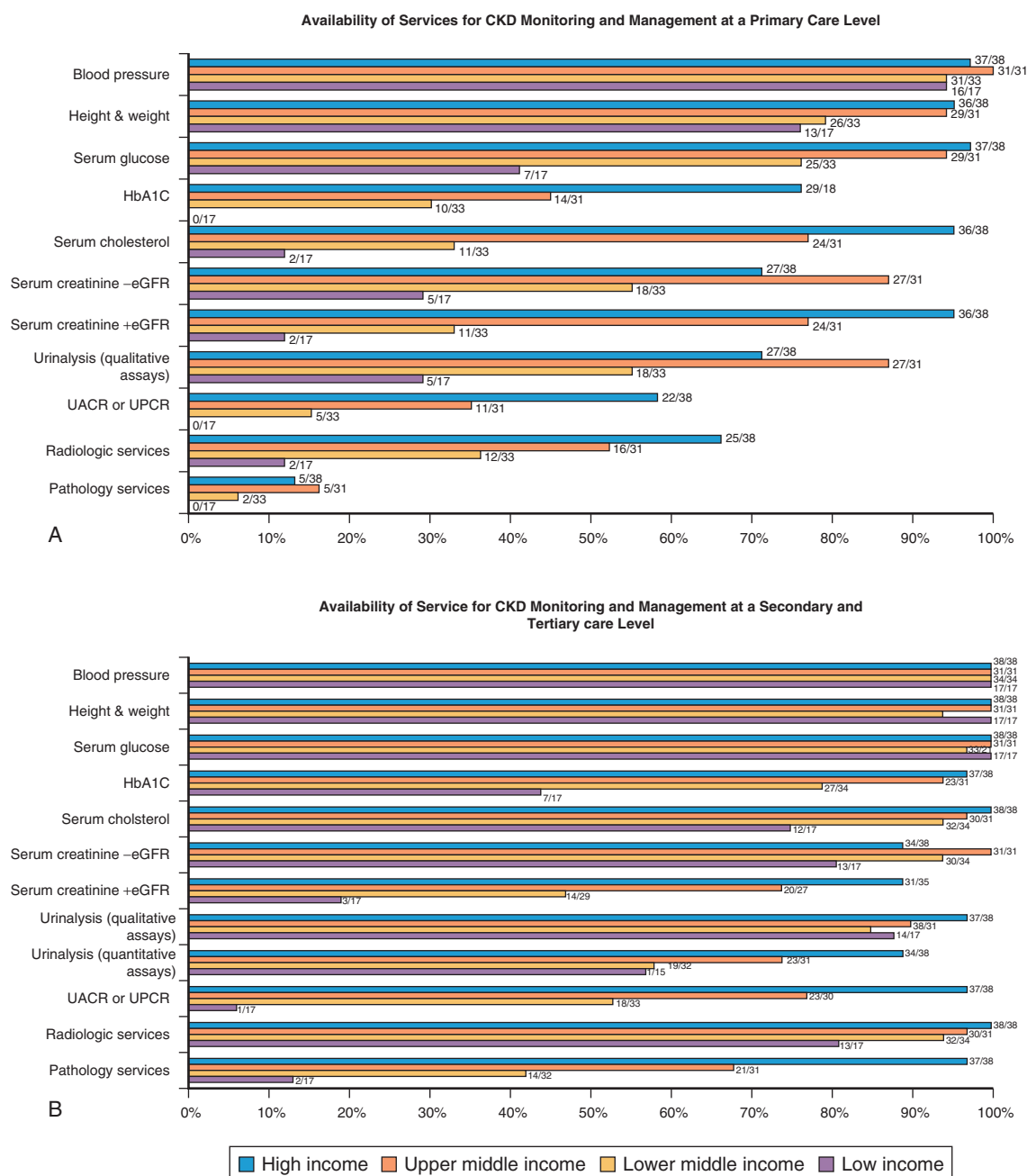


Fig. 75.14 Availability of health care services for chronic kidney disease (CKD) care at primary (A) and secondary and tertiary levels (B), dialysis (C), and kidney transplantation (D), percentage of countries with available health care service. eGFR, estimated glomerular filtration rate; HbA1C, Hemoglobin A1c; UACR, urine albumin-to-creatinine ratio; UPCR, urine protein-to-creatinine ratio. (Data from Global Kidney Health Atlas. https://www.theisn.org/images/ISN_Biennial_Report_2011-2013/GKHAAtlas_Linked_Compressed1.pdf.)

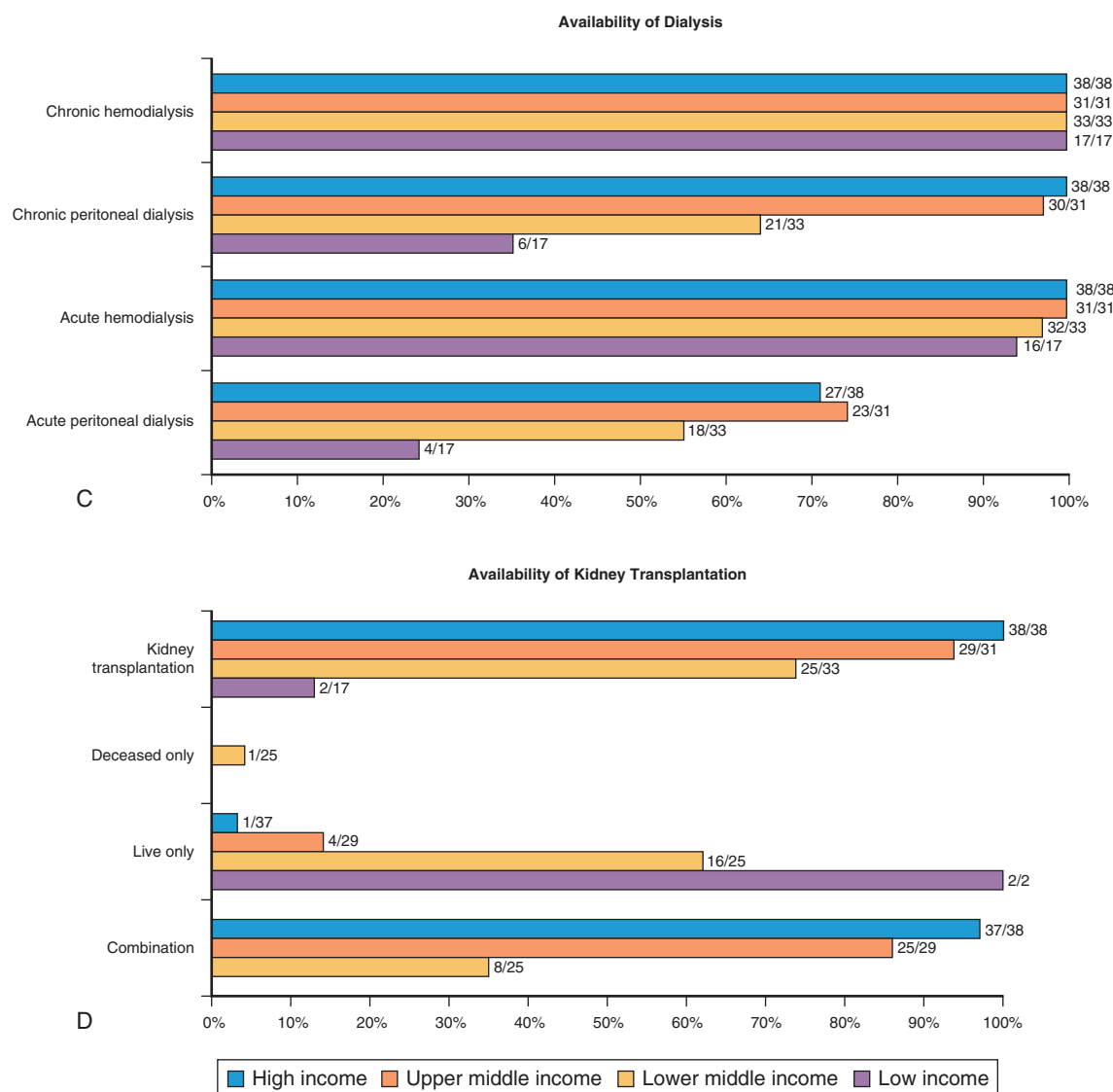


Fig. 75.14, cont'd

widely acknowledged that there is a shortage of randomized controlled trials guiding decision making in nephrology care. Enhancing capacity for and involvement in research globally, and especially in LICs, through improved funding of research programs and coordinating academic centers may lead to a more representative understanding of kidney disease globally.

The GKHA has demonstrated significant inter- and intra-regional variability in the current capacity for kidney care across the world.⁹⁵ Important gaps in awareness, services, workforce, and capacity for research were identified in many countries and regions. The findings have implications for policy development toward the establishment of robust kidney care programs, particularly for LLMICs. Basic infrastructure for services must be strengthened at the primary care level for early detection and management of AKI and CKD across all countries.^{4,32} Such information is crucial for advocacy among governmental and nongovernmental stakeholders to support countries in improving the quality of kidney care. The GKHA provides a baseline understanding of where

countries and regions stand with respect to several domains of the health system, thus allowing the monitoring of progress through the implementation of various strategies targeted toward improving kidney care. The GKHA is freely available and will be updated biannually to develop country-specific score cards and facilitate identification of country-specific gaps and targets to enhance local kidney health (<https://www.theisn.org/initiatives/ckd#health-atlas>).

PUBLIC HEALTH INITIATIVES AND POLICIES TO IMPROVE KIDNEY HEALTH AND ACCESS TO CARE

The public health approach to disease control must consider action at all levels of the health system and beyond:

- At the macro level, government health policy and inter-sectoral collaboration are key to improving population health.
- At the meso level, health administrators must translate policies into action and conduct monitoring and evaluation exercises to determine effectiveness and address problems.

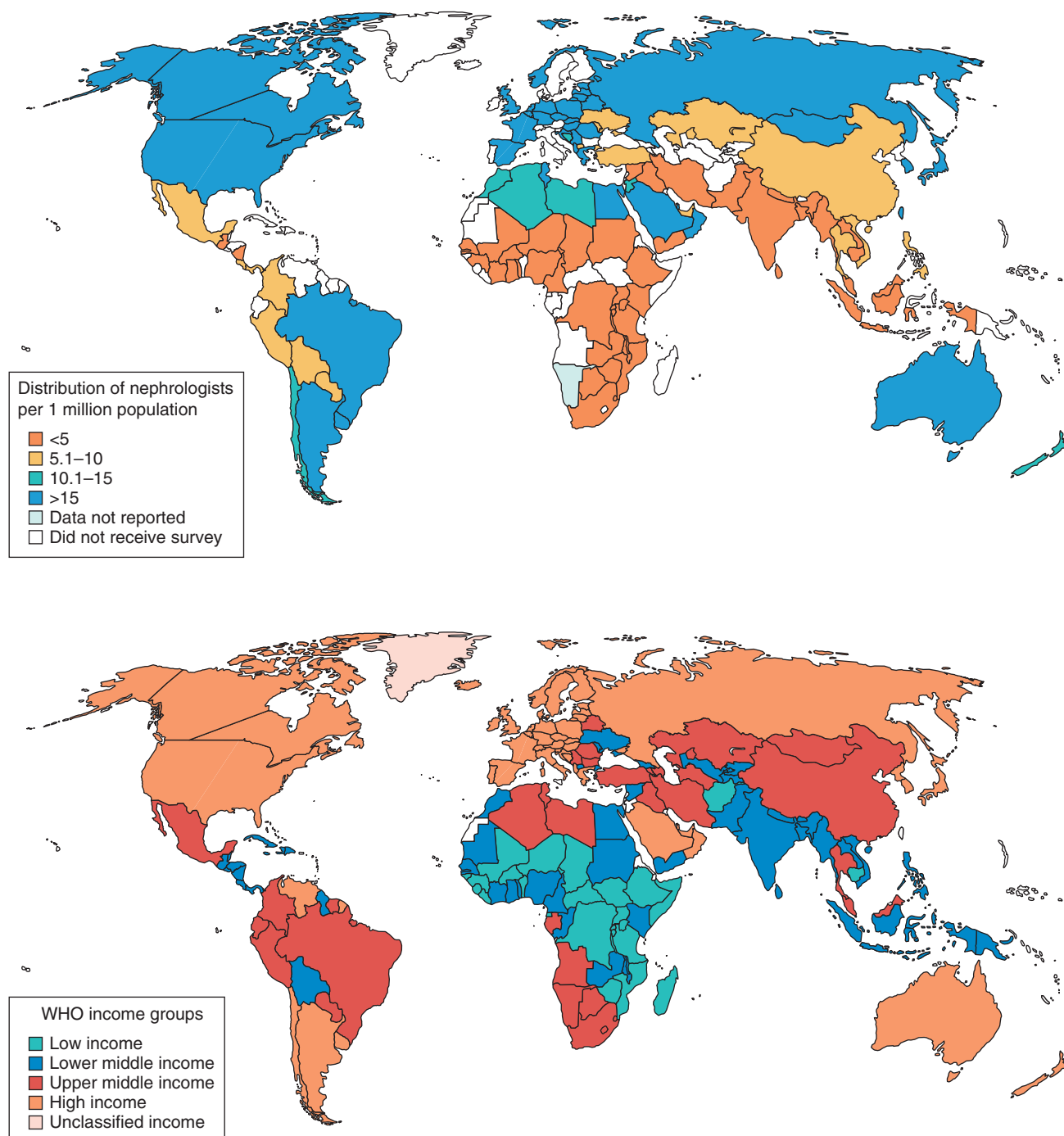


Fig. 75.15 Distribution of the global nephrologist workforce. The map depicts global distribution of nephrologists/1 million population by country and region. Data not available indicate that data were either not known or not provided on the questionnaire for countries that received the survey. (Adapted from Bello AK, Levin A, Tonelli M, et al. Assessment of global kidney health care status. *JAMA* 2017;317[18]:1864–1881.)

- At the micro level, individual health care workers and patients must work together to preserve health.

Kidney disease is affected by many social and structural factors and therefore is amenable to public health interventions (Table 75.6). Government policies that have proven effective in reducing the risk of NCDs include economic incentives to reduce prices of healthy food, taxation on unhealthy foods and beverages, school programs, including health and physical

education, food advertising restrictions and standards, increasing safe spaces for sport and recreation, media campaigns promoting smoking cessation, health warnings on cigarette packages, restriction of tobacco advertising, increased taxes on tobacco, and prohibition of smoking in public spaces.^{12,13,106,107} Implementation of public health programs such as vaccinations, health education, malaria prevention, providing access to clean water, antenatal care, ensuring safe deliveries, and optimizing child nutrition can all also reduce

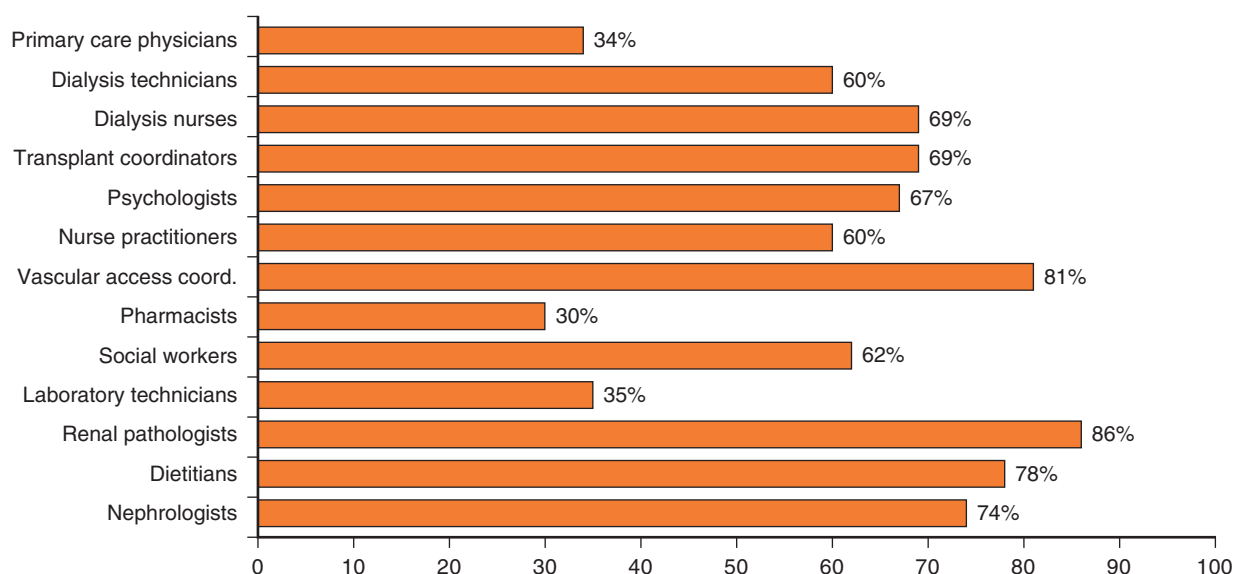


Fig. 75.16 Overall shortage in nephrology workforce. Shown are the percentages of countries globally reporting shortages of health care across the spectrum of kidney care (Data from Global kidney Health Atlas. https://www.theisn.org/images/ISN_Biennial_Report_2011-2013/GKHAAtlas_Linked_Compressed1.pdf.)

Table 75.6 Levels of Public Health Interventions, Specific Steps, and the Potential Impact on Kidney Health

Parameter	Specific Steps	Impact on Kidney Health	Comments
Socioeconomic factors	Improved livelihood, education, and sanitation; universal access to timely and affordable health care	Overall improvement in health; reduced NCD burden	Requires political will; linked to overall development and economic growth over long term; radical redesign of health systems required
Changing the context to make default decisions healthy	Reducing salt in packaged foods; higher taxation for tobacco, sweetened beverages, and processed foods; regulate OTC and indigenous medicine use Mandate protective clothing and footwear in certain occupations (e.g., outdoor exposure); access to quality reproductive care, legalization of abortions; widespread ORS availability	Reduction in kidney disease risk factors (e.g., hypertension, diabetes, metabolic syndrome); reduction in CKDnT; slowed progression of CKD; reduction in occupation-related CKD and AKI; reduction in obstetric AKI	Requires political will; likely opposition from special interest groups and industry lobby
Long-lasting protective interventions	Vaccinations, improved immunization coverage for VPDs; kidney disease screening at schools and workplaces; vector control	Reduced infection-related kidney disease (AKI and CKD); reduced infections in those with kidney disease; early prevention opportunities	Can be controversial; requires multisectoral approaches
Clinical interventions	Treatment of risk factors and predisposing conditions; measures to improve adherence to treatment	Reduction in CKD development and progression; improvement in obstetric AKI through tracking outcomes	Needs manpower and resources; ensuring a continuum of care; effective health information systems deployment
Counseling and education	Increase kidney disease awareness; dietary counseling; increase physical activity	Can slow CKD progression and prevent AKI in certain situations	Least effective; requires civil society engagement and ownership

AKI, Acute kidney injury; CKD, chronic kidney disease; CKDnT, chronic kidney disease of nontraditional causes; NCD, noncommunicable disease; ORS, oral rehydration solution; OTC, over the counter; VPD, vaccine-preventable disease.

From Joshi R, John O, Jha V, The potential impact of public health interventions in preventing kidney disease. *Semin Nephrol.* 2017;37(3):234–244.

the burden of AKI and CKD in the community.^{12,13} The risk of kidney disease begins in fetal life and spans a lifetime, so prevention of kidney disease starts with optimization of health and education for girls, universal health coverage for antenatal care and safe deliveries, and optimizing child nutrition and growth.⁸⁵

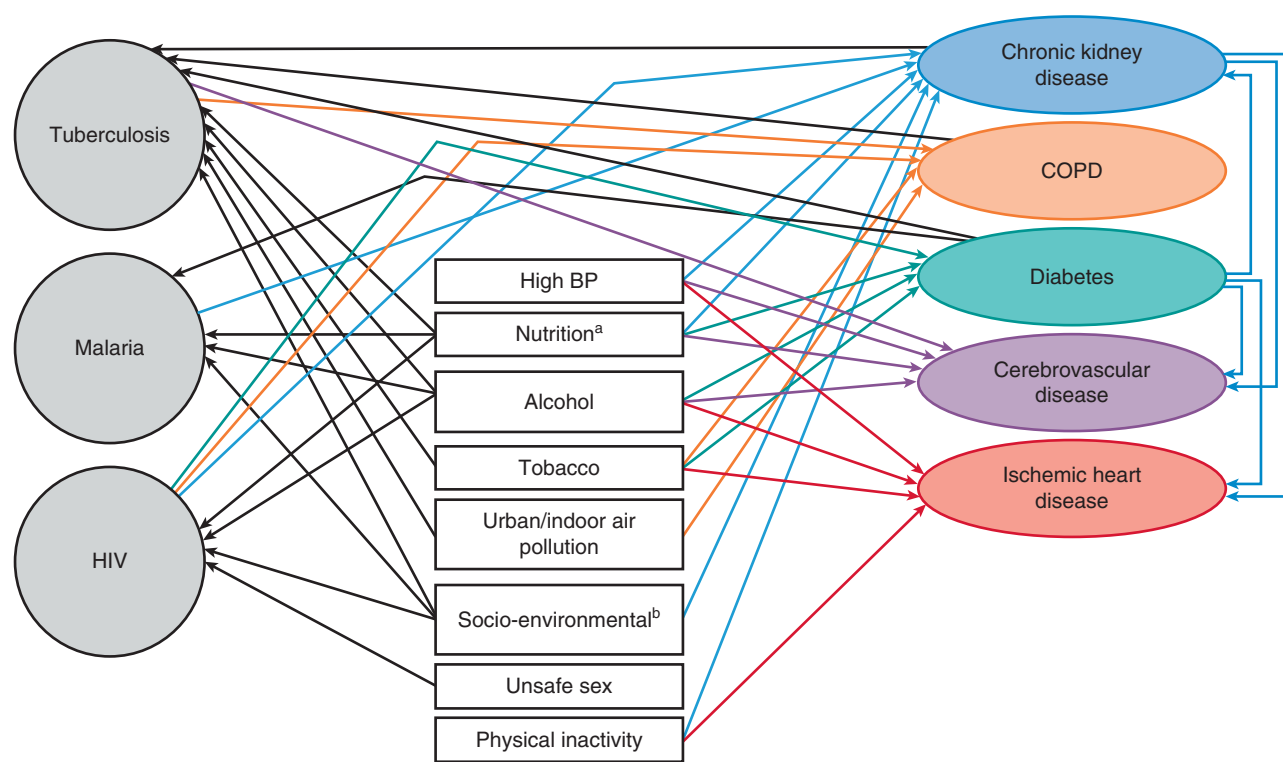
Targeted screening of high-risk individuals (with diabetes, hypertension, and CVD) has been shown to be cost-effective in detecting and treating CKD early and delaying progression to ESKD.^{29,108} Screening and early treatment strategies require availability of the necessary resources and trained personnel at all levels of the health care system to manage kidney disease.¹⁰⁹ Given the broad reach required and the need for simple screening strategies, which can be integrated into existing WHO NCD strategies, task shifting may be required and desired to improve uptake and coverage, especially in countries with very large populations, such as India.^{110–112} Systematic screening activities will also facilitate the generation of renal registries and provide information on local kidney disease burdens necessary for priority setting and program planning.¹¹³ Integration of care for patients with kidney disease into communicable and NCD programs, coupled with community engagement and empowerment of patients, promises to improve quality of care with greater efficiency and at lower

cost, as well as providing a more holistic approach to patient care (Fig. 75.17).^{31,112,114,115}

HEALTH IN ALL POLICIES AND SUSTAINABLE DEVELOPMENT GOALS

The implementation of population-level strategies to reduce the risk of NCDs and kidney disease requires coordinated multisectoral action across government and society, as emphasized in the health in all policies strategy.^{107,116,117} The health in all policies framework reflects the principles of “legitimacy, accountability, transparency and access to information, participation, sustainability and collaboration across sectors and levels of government” as important components of an equitable approach to improving population health.¹¹⁷ This multisectoral approach highlights the impact of the non-health sectors, such as industry, education, housing, transportation, agriculture, and environment, in improving population health.

In recognition of this multisectoral impact on health and the negative impact of illness on communities and overall economic growth, the United Nations launched the Sustainable Development Goals (SDGs) project in 2015 with the goal of achieving healthy people living on a healthy planet.^{78,118} Multiple social and structural factors affect the risk of kidney



^aEncompasses underweight, overweight/obesity, low fruit/vegetable consumption, high glucose intake

^bConditions associated with informality: overcrowding, unsafe water, & sanitation

Fig. 75.17 Interaction between tuberculosis (TB), malaria, and HIV and risk factors, disease precursors, and noncommunicable diseases. BP, Blood pressure; COPD, chronic obstructive pulmonary disease. (Adapted from Oni T, Unwin N. Why the communicable/non-communicable disease dichotomy is problematic for public health control strategies: implications of multimorbidity for health systems in an era of health transition. *Int Health*. 2015;7[6]:390–399.)

disease across the life course, including poverty, over- and undernutrition, poor education, access to public health interventions and health care, gender inequality, environmental and occupational factors, urban or rural location, violence, social disruptions and war, climate change, and the effects of globalization. Each SDG has direct or indirect relevance to kidney health, so progress toward each SDG promises to improve the global burden of kidney disease.¹¹⁹ It is important to acknowledge the links with other diseases; therefore, the nephrology community has the opportunity to lead in working with public health professionals and all levels of health care workers on the ground to improve synergies and coordination between disease silos to improve overall health and thereby kidney health.

HEALTH TECHNOLOGY ASSESSMENT: THE EXAMPLE OF THAILAND

As discussed earlier, reliance on OOP payments for RRT or rationing of dialysis access are highly inequitable and contribute to many annual deaths globally.^{3,71,120} The costs of dialysis and transplantation are high for all health care systems. Even in an HIC such as the United States, which otherwise does not have universal health coverage, the need to create an exception and provide coverage for ESKD patients has long been recognized and implemented (see Chapter 82).¹²¹ The challenge, however, is how to provide equitable access to RRT sustainably and without compromising funding for other health priorities. Systematic analyses of local circumstances in terms of disease burden, projected costs (direct, indirect, and opportunity costs), and expected benefits are required prior to developing policies around the provision of RRT. Such analyses are often not performed, leading to the provision of inadequate services that may exacerbate inequities.^{58,122} The example of Thailand's approach to the provision of RRT under universal health coverage is highly instructive. Thailand is an MIC that introduced universal health coverage in 2002.¹²³ At that time, universal coverage for RRT was deliberately excluded because of the high anticipated costs. This meant that patients covered under two private insurers (25% of the population) had free access to RRT, but the rest did not.¹²⁴

Over the ensuing years, policy makers conducted a comprehensive series of studies to investigate the potential demand for RRT, survey public opinion, conduct policy and cost-effectiveness analyses, and understand the impact of high OOP costs on individuals and families.¹²⁵ Specifically, it was found that many households experienced catastrophic health expenditures, with some households spending up to 70% of total income to keep one member on dialysis; often, costs were met through reduction of food consumption and dialysis frequency.¹²⁶ In the context of some favorable political circumstances, in 2008, the decision was made to include dialysis under the universal coverage scheme to reduce catastrophic expenditures and improve equity in access to dialysis across all health care schemes, despite evidence that dialysis per se is not cost-effective compared with many other interventions.¹²⁵

Prior to implementation, much consideration was given to cost containment, including a PD-first policy (analyses had shown that PD was more cost-effective than HD); bulk negotiation for pricing of dialysis supplies and medications (e.g., 25% reduction in cost of HD, 62% reduction in cost of

erythropoietin); and negotiated incentive and reimbursement strategies.^{124,125} PD centers of excellence providing training and support to peripheral centers were established, and active stakeholder, community, and patient engagement programs were conducted to enhance uptake and provide support in communities to meet additional needs, such as transport costs.¹²⁴ The numbers of patients requiring RRT initially grew rapidly as services expanded but have since plateaued, and outcomes are comparable with those of other programs.¹²⁴ Thailand is a world leader in health technology assessment, as evidenced by the deliberate and systematic approach it has taken in developing dialysis policy decisions. The Thai Health Intervention and Technology Assessment Program (HITAP; globalhitap.net) provides a framework that is adaptable and is being used by other countries in their decision making around RRT.

CONCLUSION

The global burden of kidney disease has been relatively underappreciated until rather recently. International activities are increasing to quantify this burden better, detect and address gaps in kidney care, and advocate for patients unable to access RRT.

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 Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. Prevention is the best form of treatment. In countries with limited resources, screening of which group has been shown to be cost-effective?
 - a. Screen all adults from 20 to 80 years of age.
 - b. Screen subjects with a diagnosis of hypertension and diabetes.
 - c. Screen children in primary school for low-molecular-weight proteinuria.
 - d. Screen subjects with a family history of kidney disease.
 - e. If resources permit, screen subjects who have experienced acute kidney injury or preeclampsia.

Answer: b

Rationale: Although it may seem logical to screen all those at increased risk for chronic kidney disease (CKD), at present, the data only support screening of hypertension and diabetes. This does not mean that other high-risk groups should not be screened should resources permit; however, studies are required to show cost-effectiveness. Screening all adults for CKD is not cost-effective. See reference 29.

2. Which factors should be taken into consideration when considering whether to initiate dialysis services in a low-income country?
 - a. Local epidemiology of kidney disease
 - b. Local priority health conditions
 - c. Anticipated costs of provision of dialysis per patient
 - d. Proportion of population with health insurance
 - e. Planned location of dialysis services
 - f. All of the above

Answer: f

Rationale: When planning any health intervention, it is important to gather accurate data on the prevalence and incidence of the condition in question—here, kidney

disease—and to be able to determine the population burden in comparison with other health care priorities. In countries with few resources, a priority-setting exercise is necessary to ensure best use of resources according to the three pillars of universal health coverage. It is important to determine costs up front to plan whether all who would need to be served can have access without experiencing catastrophic health expenditures. The plans should include financing sustainability, location of units for maximum access, and transparent policies governing access to care. It is also crucial to ensure that appropriate expertise is available to deliver a safe service.

3. Which public health interventions may have a direct impact in reducing the global burden of kidney disease?
 - a. Distribution of insecticide-treated nets for malaria prevention
 - b. No sale of alcohol to those younger than 21 years of age
 - c. Imposition of a sugar tax
 - d. Wearing of bicycle helmets
 - e. Improve access to antenatal care

Answer: a, c, e

Rationale: Prevention of malaria may reduce malaria-associated acute kidney injury (AKI). Improved attendance at an antenatal clinic may indicate women at risk of preeclampsia, increase safe deliveries, and reduce postpartum AKI. Imposition of a sugar tax may reduce diabetes and obesity and thereby may prevent future CKD. Alcohol and helmet legislation alone are less likely to have a large impact on reducing global AKI or CKD, although they may have indirect effects by reducing liver disease or accidents.